

Unit 1

INFORMATION REPRESENTATION

124 items · 782 marks

Syllabus sections in this unit

- 1.1 Data Representation
- 1.2 Multimedia
- 1.3 Compression

Unit 1: Information representation

1. M/J 26/P11/Q1/a,b,c,d

1 A computer can store numbers and text in binary.

(a) The computer uses the UNICODE character set.

(i) Complete the table by calculating the missing values.

character	denary	binary	hexadecimal
⋮		0001 1110 1001	1E9
@	64		40
Ω	911	0011 1000 1111	
Ψ		0001 0101 1101 0110	15D6

[4]

working space

(ii) Identify **one** other character set and explain how it is different from UNICODE.

character set

differences

[3]

(b) The computer also uses two's complement.

Convert the 8-bit two's complement binary integer 01101011 into denary.

[1]

(c) Show how the denary number 30 is subtracted from the binary number 01101011 using binary subtraction.

Show your working.

[2]

(d) Identify **two** differences between binary prefixes and decimal prefixes.

1

2

[2]

2. M/J 26/P11/Q5/a,b,c

- 5** A school needs to create a new logo. The logo can be created as a bitmap image or a vector graphic.

(a) Explain how a bitmap image is encoded in a computer.

[3]

(b) Complete the description of a vector graphic by writing the missing terms.

A is an individual element of an image, for example: a line, a square.

A is an attribute of one part of the image, for example: a line colour, the width of a line.

A is all of the elements that are included in the vector graphic.

[3]

(c) Tick (✓) **one** box to identify whether a bitmap image or a vector graphic will be most appropriate for the school logo.

Justify your choice.

	tick (✓)
bitmap image	
vector graphic	

justification

[3]

3. M/J 26/P12/Q1/a,b,c,d

- 1 (a) Tick (✓) **one** box to identify the correct definition of a pixel in a bitmap image.

- | | |
|--------------------------|---|
| ☐ | a colour in an image |
| ☐ | a row of consecutive identical colours in an image |
| ☐ | the binary code for one colour in an image |
| ☐ | the smallest element of an image that can be one colour |

[1]

- (b) Describe the difference between the terms image resolution and screen resolution.

[2]

- (c) Two items in a bitmap image file header are the resolution and colour depth.

Identify **one** other item included in the file header of a bitmap image.

[1]

- (d) The following bitmap image is created. Each letter represents a colour: R is red, W is white, B is blue.

R	R	W	W	W	W
B	B	R	R	W	W
W	W	W	R	R	R
W	W	W	W	R	R
W	W	R	R	R	R
R	R	R	R	R	B
B	B	B	B	B	B

- (i) Identify the smallest bit depth that can be used for the image shown.

[1]

- (ii) Explain how the image shown can be compressed using lossless compression.

[3]

4. M/J 26/P12/Q4/a,b

4 A sound file has a sampling rate of 44.1 kHz.

(a) Describe what is meant by a sampling rate of 44.1 kHz.

[1]

(b) A student creates a recording of a sound with a sampling resolution of 2 bytes.

The student creates a second recording of the same sound but with a sampling resolution of 8 bits.

Explain how the two sound recordings **and** the two sound files will be affected by the different sampling resolutions.

[3]

5. M/J 26/P12/Q5/a,b

5 (a) Complete the table by calculating the missing values.

denary	unsigned binary	hexadecimal	Binary Coded Decimal (BCD)
160	10100000	A0	
	100100001	121	001010001001
2486		9B6	0010010010000110
35123	1000100100110011		00110101000100100011

[4]

working space

(b) Complete the table by writing the answer to each statement.

statement	answer
The smallest 12-bit two's complement binary number	
The denary value represented by the 8-bit unsigned binary number 01100010 after a logical right shift of 2 places has been applied.	
The binary value after a cyclic left shift of 4 places has been applied to the binary number 10010101	

[3]

6. M/J 26/P13/Q3/a,b,c(i)

3 An animation is a sequence of bitmap images with a sound recording played at the same time.

(a) The sound is recorded with a sampling resolution of 2 bytes.

State what is meant by a sampling resolution of 2 bytes.

[1]

(b) (i) Explain how the data for the bitmapped images are encoded.

[3]

(ii) The bitmap images have a bit depth of 2 bytes. Each image is 2048 pixels wide by 1536 pixels high. One second of video contains eight individual bitmap images.

There is no sound recording in this part of the video.

Calculate an estimate for the file size of one second of the video in mebibytes (MiB).
Show all of your working.

working:

answer MiB

[2]

(c) The sound and images are compressed independently before being transmitted over the internet to public cloud storage.

(i) Explain the different ways that lossless compression can be used to compress the images and the sound file.

image file

sound file

[4]

7. M/J 26/P13/Q6/a,b,c,d

- 6 (a) Draw **one or more** lines to match each denary number with its equivalent unsigned binary, hexadecimal or Binary Coded Decimal (BCD) value(s).

Denary number	Equivalent unsigned value
	Hexadecimal: 121
3120	BCD: 00010101100001010001
289	Binary: 110000110000
22516	Hexadecimal: 57F4
15851	Binary: 11110111101011
	BCD: 0011000100100000

[4]

- (b) Complete the binary addition. Show all of your working.

$$\begin{array}{r}
 110101001111 \\
 + 111111001111 \\
 \hline
 \end{array}$$

[2]

- (c) Convert the two's complement integer 011010110001 into denary.

[1]

- (d) Tick (✓) **one** box to identify the smallest integer that can be represented by a 16-bit two's complement integer.

- ☐ 0000000000000000
☐ 1111111111111111
☐ 0111111111111110
☐ 1000000000000000

[1]

8. O/N 25/P11/Q1/a,b

- 1 (a) (i) Convert the binary number into hexadecimal.

101100111010

[1]

- (ii) Convert the denary number into Binary Coded Decimal (BCD).

108

[1]

- (iii) Convert the 12-bit two's complement binary integer into denary.

Show your working.

111110111100

Working

Denary value

[2]

- (b) (i) The following binary addition is performed using 8-bit registers.

Complete the calculation using binary addition.

$$\begin{array}{r} 1\ 0\ 1\ 1\ 0\ 0\ 1\ 1 \\ +\ 0\ 1\ 1\ 1\ 1\ 0\ 0\ 0 \\ \hline \end{array}$$

[1]

- (ii) Name and describe the error that can occur when binary addition is performed.

[2]

9. O/N 25/P11/Q7/d

- 7 Web browsers use Internet Protocol (IP) addresses.

- (d) A bitmap file is downloaded. The image has a maximum of 256 colours and measures 512 pixels wide by 2048 pixels high.

- (i) Calculate an estimate of the file size of the bitmap in kibibytes.

Show your working.

Working

File size of bitmap kibibytes

[2]

- (ii) Explain why the actual file size may be larger than the one calculated in (d)(i).

[2]

- (iii) Explain how a bitmap image is compressed using run-length encoding (RLE).

[2]

10. O/N 25/P12/Q6/a,b,d

- 6 (a)** A sound file is compressed by reducing the sampling rate.

State whether this is lossless or lossy compression. Justify your choice.

Type of compression

Justification

[1]

- (b)** The following table shows some words and corresponding denary values.

Word	Denary value
Computing	55
Science	56
Computers	57
are	58
Brilliant!	59
is	60
Fun!	61
Amazing!	62

The following table shows three bytes of data that have been received.

Use the table to find the corresponding words from the binary values received.

Binary value	00111000	00111100	00111110
Word			

Working

[1]

- (d)** A bitmap image has a resolution of 1000 pixels wide by 2000 pixels high. The colour depth is 16 bits.

Calculate an estimate of the file size in megabytes.

Show your working.

File size megabytes

[2]

11. O/N 25/P12/Q7/a,b,c

- 7 (a) The following binary addition is performed using 8-bit registers.

Complete the calculation using binary addition.

$$\begin{array}{r} 1\ 0\ 0\ 0\ 0\ 1\ 1\ 1 \\ +\ 0\ 0\ 1\ 1\ 1\ 0\ 0\ 1 \\ \hline \end{array}$$

[1]

- (b) A computer uses the Unicode character set.

State the number of bits used to store **one** character from the Unicode character set.

[1]

- (c) ASCII is another character set. The ASCII value for the character 'h' has the denary value 104.

- (i) Write the Binary Coded Decimal (BCD) value for the ASCII character 'h'.

[1]

- (ii) Write the hexadecimal value for the ASCII character 'h'.

[1]

12. O/N 25/P13/Q1/a,b,c

- 1 (a) A student creates sound recordings.

Making changes to a sound recording may impact the accuracy of the resulting sound file or the size of the file.

Draw **two** lines from each change to the impacts it has on the sound file.

Change	Impact
increase the duration of the recording	the file size gets bigger
	no change in the file size
increase the sampling rate	the file size gets smaller
	the accuracy of the sound file improves
decrease the sampling resolution	no change to the accuracy of the sound file
	the accuracy of the sound file worsens

[3]

- (b) The file names of the sound recordings are stored using the ASCII character set.

Explain how text is represented by the ASCII character set.

[2]

- (c) Give **two** differences between the ASCII and UNICODE character sets.

1

2

[2]

13. O/N 25/P13/Q2/a,b,d

- 2 (a)** Data in a computer system is represented in binary.

Put **one** tick (✓) in each row to identify the minimum number of bits used to store each example of data.

Example of data	Number of bits						
	4	8	16	24	32	64	128
the hexadecimal value F139							
16 000 000 unique amplitude values							
an IPv4 address							
256 unique colours							
an IPv6 address							
the denary value 65 000							

[3]

- (b)** Convert the denary number –108 into a 12-bit two's complement binary number.

[1]

- (d)** Convert the following positive binary integer into hexadecimal.

1110001100111011

[1]

14. O/N 25/P13/Q7/b

- 7** A digital video camera records students in a classroom. The data is transferred from the digital video camera over the internet to a server. Artificial Intelligence (AI) is used to analyse the video on the server to identify when students are interacting with the lesson and the teacher.

- (b)** A video is made of many bitmap images called frames. 30 frames are recorded every second.

Each frame is 4000 pixels wide by 3000 pixels high. The video records using 16-bit colour depth.

Calculate an estimate for the file size for **one** second of the video in gigabytes.

Show your working.

File size gigabytes

[2]

15. M/J 25/P11/Q2/b

- 2** Programmers in a software development company take part in live video conferences to discuss their work.

The live video conferences take place using real-time bit streaming. The video is compressed before it is transmitted.

- (b) (i)** Explain the reasons why a video is compressed before it is transmitted using real-time bit streaming.

[4]

- (ii)** Identify whether the lossy or lossless compression method is more appropriate for real-time bit streaming. Justify your answer.

Compression method

Justification

[3]

16. M/J 25/P11/Q3/a,b

3 A computer stores images and text files.

(a) One of the images is a bitmapped image.

Complete the table by writing the answer for each statement.

Statement	Answer
the term for the smallest element that makes up an image	
the largest number of different colours that can be represented with a bit depth of 8 bits	
the term for the dots per inch (dpi) when an image is displayed	

[3]

(b) The text in the files is stored using the Unicode character set.

(i) Give **two** advantages of using the Unicode character set instead of using the ASCII character set.

1

2

[2]

(ii) The Unicode character 'Ƨ' has the binary value: 0010 0111 0110 1110

Convert the binary value for the character 'Ƨ' into denary.

[1]

(iii) The Unicode character 'Σ' has the hexadecimal value 2140

Convert the hexadecimal code for the character 'Σ' into denary.

[1]

17. M/J 25/P12/Q2/a,b,c

- 2 (a) Convert the denary integer 558 into 12-bit binary and hexadecimal.

Binary

Hexadecimal

[2]

- (b) (i) Convert the two's complement binary integer 11100010 into denary.

[1]

- (ii) Write the smallest and the largest two's complement binary integers that can be represented in 8 bits.

Smallest

Largest

[2]

- (c) Give **one** application where Binary Coded Decimal (BCD) is used and justify its use.

Application

Justification

[2]

18. M/J 25/P13/Q1/a,b,c

- 1 (a) A student paints a picture in an art class.

The student takes a photograph of the picture using a digital camera. The digital camera creates an image with a resolution of 2 million pixels and uses a bit depth of 16 bits.

- (i) Calculate the file size of the image created by the digital camera in megabytes (MB).

Show your working.

Working space

Answer MB

[2]

- (ii) The student takes a second photograph with a lower bit depth.

Explain the effect of decreasing the bit depth on the image and on the image file.

Effect on image

Effect on image file

[4]

- (b) A teacher records an audio file of the student playing the piano during a music lesson.

Complete the table by giving the term for each description about sound representation on a computer.

Description	Term
the number of times the amplitude is measured per time interval	
the number of bits used to store each amplitude measurement	
the type of sound wave before it is recorded by a computer	

[3]

(c) The student types a report on a computer in a history lesson.

(i) The computer uses the Unicode character set.

Give **two** characteristics of the Unicode character set.

1

2

[2]

(ii) Some of the school's computers use the extended ASCII character set.

The table gives some characters from the extended ASCII character set, their denary, 8-bit binary and hexadecimal numbers.

Complete the table by filling in the missing numbers.

Character	Denary	8-bit Binary	Hexadecimal
!	33		21
L		01001100	4C
ü	252	11111100	

[3]

19. M/J 25/P13/Q7/b,c

(b) Complete the following binary addition. Show your working. Include any overflow bit(s).

$$\begin{array}{r}
 1 \ 1 \ 1 \ 1 \ 0 \ 1 \ 0 \ 1 \\
 + \ 1 \ 0 \ 1 \ 1 \ 0 \ 0 \ 0 \ 1 \\
 \hline
 \end{array}$$

[1]

(c) Subtract the binary number 00011110 from the binary number 01100100 using binary subtraction. Show your working.

[2]

20. O/N 24/P11/Q1/a,b,c,d

- 1 (a) State **one** difference between a tebibyte and a gigabyte.

[1]

- (b) (i) Convert the unsigned binary integer into hexadecimal.

110001100111

Answer

[1]

- (ii) Convert the two's complement binary number into denary.

100110010111

Answer

[1]

- (iii) Convert the Binary Coded Decimal (BCD) into denary.

010101110011

Answer

[1]

- (c) Subtract the denary number 23 from the two's complement binary number 01001010

Perform this calculation using binary subtraction.

Show your working.

Working

Answer

[2]

- (d) State **one** reason why binary addition and subtraction can result in overflow.

[1]

21. O/N 24/P11/Q4/b

- 4 A student uses a laptop to write a program that is saved as a text file.

- (b) The student compresses the file before it is emailed to their teacher as an attachment.

- (i) Explain the benefits to the teacher of the attachment being a compressed file.

[3]

- (ii) Describe **one** lossless method of compressing a text file.

[3]

22. O/N 24/P12/Q7/a,b

7 A student takes a photograph of a science experiment.

(a) The photograph is saved as a bitmapped image.

(i) Define the following bitmap terms.

Colour depth

File header

[2]

(ii) Explain why changing the image resolution will affect the image quality and file size.

Image quality

File size

[2]

(iii) Identify **one** lossless method of compressing an image.

[1]

(b) The student draws a picture on paper that is scanned into the computer and saved as a vector graphic.

Define the vector graphic terms property and drawing list.

Property

Drawing list

[2]

23. O/N 24/P13/Q6/a,b,c

6 A computer system stores text, images and sound.

(a) A character set is used to represent characters in a computer.

Identify **and** describe **one** character set.

Character set

Description

[2]

(b) The colour of each pixel in a bitmapped image is represented by 8 bits.

(i) State the largest number of different colours that can be represented by 8 bits.

[1]

(ii) State **one** drawback of increasing the number of bits that represents each pixel in the bitmap image.

[1]

(iii) A bitmap image can be compressed using lossy compression.

Explain the reasons why lossy compression is often suitable for a bitmap image.

[2]

(c) (i) Explain how an analogue sound wave is converted into digital data.

[2]

(ii) Describe **one** method of compressing a sound file using lossy compression.

[2]

24. O/N 24/P13/Q8/a,b

8 (a) Convert the hexadecimal number 1FAB into denary.

Working

Denary value

[1]

(b) Explain how to convert the two's complement integer 10011111 into denary. Give the denary value after conversion.

Explanation

Denary value

[3]

25. M/J 24/P11/Q2/d

- 2 A video doorbell is attached to the front door of a house. The doorbell uses a motion sensor to detect when a visitor walks in front of the door. When the motion sensor is activated:
- The digital camera in the doorbell starts recording a video.
 - A message is transmitted to a smartphone so that the person who lives in the house can watch the video.

The doorbell also has a button that can be pressed. When the button is pressed, a message is transmitted to a smartphone to play the doorbell sound.

The videos are stored on the doorbell's internal secondary storage device and overwritten when the secondary storage device is full.



- (d) The digital camera has a microphone which is used to record the sound for the video.

The user changes the sampling rate that the microphone uses from 44.1 kHz to 88.2 kHz.

Describe how this change in sampling rate will affect the performance of the video doorbell.

[3]

26. M/J 24/P11/Q7

- 7 Complete the binary addition. Show your working.

$$\begin{array}{r} 10011110 \\ 01100001 \\ + 00011001 \\ \hline \end{array}$$

[3]

27. M/J 24/P12/Q2/d

- 2** A computer game is being designed that users will be able to play using a virtual reality (VR) headset.

(d) The computer can transmit a video made from bitmap images and vector graphic animations to the VR headset.

(i) Describe how the data for a bitmapped image is encoded.

[3]

(ii) Describe the contents of a vector graphic drawing list.

[2]

(iii) The bitmap video is **not** compressed before transmission to the VR headset.

Give **two** reasons why the video does **not** need to be compressed.

1

2

[2]

28. M/J 24/P12/Q7/a,b,c

- 7** A computer stores binary data.

(a) Tick (✓) **one** box only to identify the **largest** file size.

- | | |
|--------------------------|----------------|
| ☐ | 3300 kibibytes |
| ☐ | 0.3 megabytes |
| ☐ | 3 mebibytes |
| ☐ | 3300 kilobytes |

[1]

(b) Subtract the denary number 10 from the denary number 100 using binary subtraction.

Show your working.

Working

Answer

[3]

(c) Convert the hexadecimal number C0F into denary.

Show your working.

Working

Answer

[2]

29. M/J 24/P13/Q1/a,b,c,d

1 (a) Complete the following description.

A kibibyte has a prefix. Three kibibytes is the same as bytes.

A megabyte has a prefix. Two terabytes is the same as gigabytes.

[4]

(b) Convert the denary number 241 to hexadecimal.

[1]

(c) State what is meant by an **overflow in binary addition**.

[1]

(d) Computers use character sets when representing characters in binary.

(i) Complete the table by identifying the number of bits each of the character sets allocates to each character.

Character set	Number of bits
ASCII	
extended ASCII	
Unicode	

[1]

(ii) Explain how the word 'Clock' is represented by a character set.

[2]



30. M/J 24/P13/Q2/a,b,c

2 A photograph is stored as a bitmap image.

- (a) The photograph has a resolution of 4000 pixels wide by 3000 pixels high. The bit depth is 4 bytes.

Calculate an estimate for the file size of the photograph in megabytes.

Show your working.

Working

Answer megabytes

[2]

- (b) The photograph is compressed before being uploaded to a web server.

- (i) Give **three** benefits of this photograph being compressed using lossy compression instead of lossless compression.

1

2

3

[3]

- (ii) Explain how run-length encoding (RLE) will compress the photograph.

[2]

- (c) Identify **two** elements of a bitmap image that can be changed to reduce its file size.

1

2

[2]



31. O/N 23/P11/Q1/a,b,c

- 1 (a) Draw **one** line from each vector graphic term to its most appropriate description.

Term	Description
drawing list	a component created using a formula
drawing object	defines one characteristic of a component
property	data required to create all components in the graphic

[2]

- (b) State what is meant by the **bit depth** of a bitmap image **and** explain how changing the bit depth affects the image.

Definition

Explanation

[3]

- (c) Explain why a bitmap image is often compressed before it is attached to an email.

[2]

32. O/N 23/P12/Q3/a,b,c

- 3 (a) State **one** difference between a kibibyte and a megabyte.

[1]

- (b) (i) Convert the denary value into a 12-bit two's complement binary integer.

–196

Answer

[1]

- (ii) Convert the Binary Coded Decimal (BCD) into denary.

100001100101

Answer

[1]

- (iii) Convert the unsigned binary integer into denary.

000111010110

Answer

[1]

- (c) Identify **one** practical application of BCD **and** justify why BCD is used in this application.

Application

Justification

[2]

33. O/N 23/P12/Q6/a,b,c

- 6 (a) A real-time video of a music concert needs to be streamed to subscribers.

Tick (✓) **one** box to identify the most appropriate type of compression **and** justify your answer.

Lossy	Lossless
☐	☐

Justification

[3]

- (b) Explain the impact of changing the sampling resolution on the accuracy of a sound recording.

[3]

- (c) A bitmap image has a resolution of 2048 pixels wide and 1024 pixels high.

The image has a bit depth of 10 bits per pixel.

Estimate the file size of the bitmap image in mebibytes. Show your working.

Working

Estimated file size in mebibytes

[2]

34. O/N 23/P13/Q1/a,b

- 1 (a) State what is meant by **analogue data**.

[1]

- (b) Draw **one** line from each term to its most appropriate description.

Term	Description
sampling	the number of samples taken per second
sampling rate	taking measurements at regular intervals and storing the values
sampling resolution	the number of bits used to store each sample

[2]

35. O/N 23/P13/Q2/a,b

- 2 (a) Describe the impact of increasing the image resolution on the quality of a bitmap graphic.

[2]

- (b) Calculate the file size of a bitmap image using the following information:

- image resolution of 2048 pixels wide and 1024 pixels high
- bit depth of 16 bits.

Give your answer in kibibytes. Show your working.

Working

Answer in kibibytes

[2]

36. O/N 23/P13/Q5/b(i)

- 5 A programmer uses an Integrated Development Environment (IDE) to develop a program that monitors air quality.

- (b) The program is distributed by downloading the source code and its library files from a web server.

- (i) Explain the reasons for compressing the files.

[2]

37. M/J 23/P11/Q1/a,b,c

- 1 Images are being created to advertise holidays.

Some of the images are bitmap images and some are vector graphics.

(a) Complete the table by defining the image terms.

Term	Definition
Drawing list	
Pixel	
Colour depth	

[3]

(b) The bitmap images are photographs of the holiday locations.

- (i) Colour depth and image resolution are both included in the file header of a bitmap image.

Identify **two other** items that could be included in the file header of each photograph.

1

2

[2]

- (ii) One of the photographs has a bit depth of 8 bytes and an image resolution of 1500 pixels wide and 3000 pixels high.

Calculate the file size of the photograph in megabytes. Show your working.

Working

File size MB

[2]

(c) The photographs are compressed before they are uploaded to a web server. Customers download the photographs from this web server.

- (i) Explain the reasons why compressing the photographs will benefit the customers.

[3]

- (ii) An image can be compressed using run-length encoding (RLE).

Explain the reasons why RLE may **not** reduce the file size of a bitmap image.
Give **one** example in your answer.

[3]

38. M/J 23/P11/Q3/d(i,ii,iii,iv,v)

- 3** A computer has an Operating System (OS).

- (d) The computer stores data in binary form.

- (i) State the difference between a kibibyte and a kilobyte.

[1]

- (ii) Convert the denary number 964 into Binary Coded Decimal (BCD).

[1]

- (iii) Convert the positive binary integer 11110010 into hexadecimal.

[1]

- (iv) Give the smallest **and** largest two's complement binary number that can be represented using 8 bits.

Smallest

Largest

[2]

- (v) Add the following two binary integers using binary addition. Show your working.

$$\begin{array}{r} 1\ 0\ 1\ 1\ 0\ 0\ 0\ 0 \\ +\ 0\ 0\ 0\ 1\ 1\ 0\ 1\ 1 \\ \hline \end{array}$$

[2]

39. M/J 23/P12/Q4/a,b,c

4 Data in a computer is stored in binary form.

(a) State the number of unique binary values that can be represented in 16 bits.

[1]

(b) Give the 8-bit one's complement representation of the denary number -120 .

Show your working.

Working

Answer

[2]

(c) Convert the hexadecimal number A04 into denary.

Show your working.

Working

Answer

[2]

40. M/J 23/P13/Q3/c

3 A mobile telephone is used to record a video.

(c) The video includes a sound recording.

(i) Describe how sound is represented in a computer.

[3]

(ii) A second video is recorded. The sound in the second video needs to be more precise.

Explain the reasons why increasing the sampling rate and the sampling resolution will improve the precision of the second recording.

Sampling rate

Sampling resolution

[4]

41. M/J 23/P13/Q7/a

7 A computer stores data in binary form.

(a) Draw **one** line from each description to its matching denary value.

Description	Denary value
	-127
The smallest integer that can be represented in 8-bit two's complement.	127
	-255
The largest integer that can be represented in 8-bit two's complement.	-128
	-256
The largest unsigned integer that can be represented in 8 bits.	256
	128
	255

[3]

42. O/N 22/P11/Q1/a,b,c,d

- 1 (a) (i) Convert the unsigned binary integer into denary.

00100111

Answer

[1]

- (ii) Convert the Binary Coded Decimal (BCD) into denary.

00100111

Answer

[1]

- (iii) Convert the 8-bit two's complement binary integer into denary.

11100111

Answer

[1]

- (b) Perform the following binary subtraction. Show your working.

$$\begin{array}{r} 1 \ 0 \ 1 \ 1 \ 0 \ 0 \ 1 \ 1 \\ - \ 0 \ 1 \ 1 \ 1 \ 0 \ 1 \ 0 \ 1 \\ \hline \end{array}$$

[2]

- (c) Give **one** similarity and **two** differences between the ASCII and Unicode character sets.

Similarity

Difference 1

Difference 2

[3]

- (d) Sound samples are recorded and saved in a file.

- (i) State what is meant by **sampling rate**.

[1]

- (ii) Explain the effect of increasing the **sampling resolution** on the sound file.

[2]

43. O/N 22/P11/Q7/c

- (c) An Operating System may include a utility program to compress text files.

Describe **one** appropriate method of compressing a text file.

[3]

44. O/N 22/P12/Q2/a,b

- 2 (a) (i) Convert the two's complement binary integer into denary.

10010110

Answer [1]

- (ii) Convert the unsigned binary integer into hexadecimal.

10010110

Answer [1]

- (iii) Convert the unsigned binary integer into Binary Coded Decimal (BCD). Show your working.

10010101

Working

Answer [2]

- (b) Perform the following binary addition.

1 0 0 0 1 1 0 0
+ 0 1 0 0 0 1 1 0

[1]

45. O/N 22/P12/Q6/b

- 6 (a) A student uses a networked laptop computer to send an email to a colleague.

- (b) The student records a sound file.

- (i) Explain the effect of increasing the sampling rate on the accuracy of the sound recording.

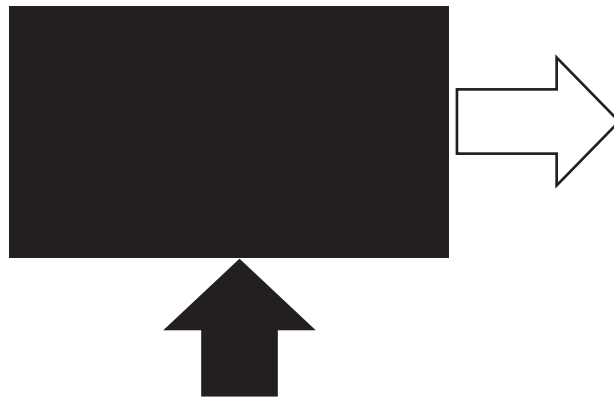
[2]

- (ii) Explain the effect of decreasing the sampling resolution on the file size of the sound recording.

[2]

46. O/N 22/P12/Q8/a,b,c

- 8 The following bitmap image has a resolution of 4096×4096 pixels and a colour depth of 24 bits per pixel.



The image is displayed on a monitor that has a screen resolution of 1920×1080 pixels.

(a) Tick (✓) **one** box in each row to identify the effect of each action on the image file size.

Action	Increases the file size	Decreases the file size	No change to the file size
Change the colour depth of the image file to 16 bits per pixel.			
Change the screen resolution to 1366×768 pixels.			
Change the colour of the rectangle from black to red.			

[2]

(b) State **two** benefits of creating a vector graphic instead of a bitmap image.

1

2

[2]

(c) A second bitmap image is stored using a colour depth of 8 bits per pixel.

The file is compressed using run-length encoding (RLE).

(i) The table shows the compressed and uncompressed values for parts of the image file.

Each colour of the pixel in the image is represented by a hexadecimal value.

Complete the table. The first row has been completed for you.

Uncompressed image	RLE compressed image
EA F1 F1 F2 F2 F2 EA	1EA 2F1 3F2 1EA
	2AB 2FF 11D 167
32 32 80 81 81	

[2]

(ii) RLE is an example of lossless compression.

Explain why lossless compression is more appropriate than lossy compression for a text file.

[2]

47. O/N 22/P13/Q1/a,b

1 A digital audio message needs to be recorded.

(a) Tick (✓) **one** box in each row to identify the effect of each action on the accuracy of the recording.

Action	Accuracy increases	Accuracy decreases	Accuracy does not change
Change the sampling rate from 40 kHz to 60 kHz.			
Change the duration of the recording from 20 minutes to 40 minutes.			
Change the sampling resolution from 24 bits to 16 bits.			

[2]

(b) The audio message is recorded with a sampling rate of 50 kHz and a sampling resolution of 16 bits.

The recording is 20 minutes in length.

Calculate the file size of the recording.

Give your answer in megabytes **and** show your working.

Working

Answer megabytes

[2]

48. O/N 22/P13/Q2/e

- 2 The relational database `ASTRONOMY` is used to store data about telescopes, the companies that own the telescopes and the photographs taken by the telescopes.

The database has these three tables:

`COMPANY (TelephoneNumber, CompanyID, CompanyName)`

`PHOTOGRAPH (PhotoID, TelescopeID, DateTaken, TimeTaken, Elevation)`

`TELESCOPE (TelescopeID, CompanyID, SerialNumber)`

- (e) The photographs are stored as bitmap images.

Complete the statements about bitmap images by writing the missing words.

The of a bitmap image is the number of bits that are used to store each pixel.

Metadata about the image is stored in the of the file.

[2]

49. O/N 22/P13/Q9/a,b

- 9 (a) (i) Convert the unsigned binary value into hexadecimal.

10010011

Answer

[1]

- (ii) Convert the unsigned binary value into denary.

10010011

Answer

[1]

- (b) State **two** benefits of using Binary Coded Decimal (BCD) to represent values.

Benefit 1

Benefit 2

[2]

50. O/N 21/P11/Q1/a,b,c

- 1 (a) Draw **one** line from each binary value to its equivalent (same) value on the right.

Binary value

8 bits

8000 bits

1000 kilobytes

1024 mebibytes

8192 bits

1 kibibyte

1 gigabyte

1 byte

1 kilobyte

1 gibibyte

1 megabyte

1 mebibyte

[5]

- (b) (i) Perform the following binary addition. Show your working.

$$\begin{array}{r} 10101010 \\ + 00110111 \\ \hline \end{array}$$

[2]

- (ii) State how an overflow can occur when adding two binary integers.

[1]

- (c) Convert the hexadecimal value F0 into denary.

[1]

51. O/N 21/P11/Q7/a,b

7 Bobby is recording a sound file for his school project.

(a) He repeats the recording of the sound several times, with a different sample rate each time.

(i) Describe the reasons why the sound is closer to the original when a higher sample rate is used.

[2]

(ii) Describe the reasons why the sound file size increases when a higher sample rate is used.

[2]

(b) Bobby wants to email the sound file to his school email address. He compresses the file before sending the email.

(i) Explain the reasons why Bobby compresses the sound file.

[2]

(ii) Bobby uses lossless compression.

Describe how lossless compression can compress the sound file.

[2]

52. O/N 21/P12/Q4/a,b,c,d,e

4 A register stores the following binary number:

1	1	0	0	1	1	0	1
---	---	---	---	---	---	---	---

(a) The binary value in the register represents an unsigned binary integer.

Convert the unsigned binary integer into denary.

[1]

(b) The binary value in the register represents a two's complement binary integer.

Convert the two's complement binary integer into denary.

[1]

(c) The binary value in the register represents a hexadecimal number.

Convert the binary number into hexadecimal.

[1]

(d) State why the value in the register cannot be interpreted as a Binary Coded Decimal (BCD).

[1]

(e) The binary contents of **two** registers are:

Register 1	0	0	1	1	1	1	0	1
Register 2	0	0	1	0	1	1	0	1

(i) Add the contents of **Register 1** and **Register 2**. Show your working.

Answer

[2]

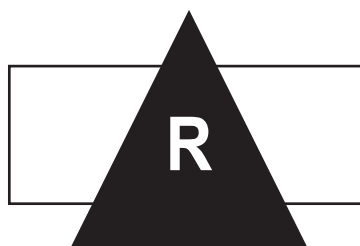
(ii) Subtract the contents of **Register 2** from the contents of **Register 1**. Show your working.

Answer

[2]

53. O/N 21/P12/Q5/a,b

5 Riya has created the following logo as a vector graphic.



- (a) Complete the table by writing a description of each vector graphic term **and** give an example for this logo.

Term	Description	Example from logo
Property		
Drawing list		

[4]

- (b) Riya takes a photograph using a digital camera. The photograph is stored as a bitmap image.

- (i) Describe **two** differences between a vector graphic and a bitmap image.

1

2

[4]

- (ii) Riya needs to email the photograph. She compresses the photograph before sending it using an email.

Describe **two** lossy methods that Riya can use to compress the image.

Method 1

Method 2

[4]

54. O/N 21/P13/Q1/a,b,c

- 1 (a) Draw **one** line from each binary value to its equivalent (same) value on the right.

Binary value

8 bits

8000 bits

1000 kilobytes

1024 mebibytes

8192 bits

1 kibibyte

1 gigabyte

1 byte

1 kilobyte

1 gibibyte

1 megabyte

1 mebibyte

[5]

- (b) (i) Perform the following binary addition. Show your working.

$$\begin{array}{r} 10101010 \\ + 00110111 \\ \hline \end{array}$$

[2]

- (ii) State how an overflow can occur when adding two binary integers.

[1]

- (c) Convert the hexadecimal value F0 into denary.

[1]

55. O/N 21/P13/Q7/a,b

7 Bobby is recording a sound file for his school project.

(a) He repeats the recording of the sound several times, with a different sample rate each time.

(i) Describe the reasons why the sound is closer to the original when a higher sample rate is used.

[2]

(ii) Describe the reasons why the sound file size increases when a higher sample rate is used.

[2]

(b) Bobby wants to email the sound file to his school email address. He compresses the file before sending the email.

(i) Explain the reasons why Bobby compresses the sound file.

[2]

(ii) Bobby uses lossless compression.

Describe how lossless compression can compress the sound file.

[2]

56. 9608 O/N 21/P11/Q1/a,b,c

1 An image can be either a bitmap image or a vector graphic.

(a) Vector graphics are made up of drawing objects and their properties.

(i) State what is meant by a **drawing object**.

[1]

(ii) Identify **four** properties of a **drawing object**.

1

2

3

4

[4]

(b) Identify **three** items that are stored in a **bitmap** file header.

1

2

3

[3]

(c) A bitmap image needs to be compressed before it can be sent by email.

Describe **one** lossy and **one** lossless method of compressing the image.

Lossy

Lossless

[4]

57. 9608 O/N 21/P11/Q2/a

2 Joanne wants to record sound files and videos for uploading to a social media website.

(a) The following table contains terms about sound representation and encoding.

Complete the table by writing the definitions for each term.

Term	Definition
Sampling	
Sampling resolution	
Sampling rate	

[3]

58. M/J 21/P11/Q1/a,b,c

1 Anya scans an image into her computer for a school project.

(a) The scanned image is a bitmapped image.

(i) Complete the following table to describe the two terms about graphics.

Term	Description
Pixel	
File header	

[2]

- (ii) The image is scanned with an image resolution of 1024×512 pixels, and a colour depth of 8 bits per pixel.

Calculate an estimate for the file size, giving your answer in mebibytes. Show your working.

Working

Answer mebibytes

[3]

- (b) The image is compressed using lossless compression.

Identify **one** method of lossless compression that can be used to compress the image **and** describe how the method will reduce the file size.

Lossless compression method

Description

[3]

- (c) One of the colours used in the image has the hexadecimal colour code:

#FC238A

FC is the amount of red, 23 is the amount of green and 8A is the amount of blue in the colour.

- (i) Convert the hexadecimal code FC into denary.

[1]

- (ii) The amount of green in binary is 00100011. This has the denary number 15 added to it to create a second colour.

Add the denary number 15 to the binary number 00100011 and give your answer in binary.

Perform the addition in binary. Show your working.

Working

Answer (in binary)

[3]

- (iii) Hexadecimal 23 in two's complement representation is 00100011. The denary number 10 needs to be subtracted from this value.

Subtract the denary number 10 from the two's complement representation 00100011.

Give your answer in binary. Show your working.

Working

Answer (in binary)

[3]

59. M/J 21/P12/Q4/b(i)

- 4 The table shows part of the instruction set for a processor. The processor has one general purpose register, the Accumulator (ACC), and an Index Register (IX).

Instruction		Explanation
Opcode	Operand	
LDM	#n	Immediate addressing. Load the number n to ACC
LDD	<address>	Direct addressing. Load the contents of the location at the given address to ACC
STO	<address>	Store contents of ACC at the given address
ADD	<address>	Add the contents of the given address to the ACC
INC	<register>	Add 1 to the contents of the register (ACC or IX)
DEC	<register>	Subtract 1 from the contents of the register (ACC or IX)
CMP	<address>	Compare the contents of ACC with the contents of <address>
JPE	<address>	Following a compare instruction, jump to <address> if the compare was True
JPN	<address>	Following a compare instruction, jump to <address> if the compare was False
JMP	<address>	Jump to the given address
IN		Key in a character and store its ASCII value in ACC
OUT		Output to the screen the character whose ASCII value is stored in ACC
END		Return control to the operating system

denotes a denary number, e.g. #123

The current contents of the main memory and selected values from the ASCII character set are:

Address Instruction

70	IN
71	CMP 100
72	JPE 80
73	CMP 101
74	JPE 76
75	JMP 80
76	LDD 102
77	INC ACC
78	STO 102
79	JMP 70
80	LDD 102
81	DEC ACC
82	STO 102
83	JMP 70
...	
100	68
101	65
102	100

ASCII code table (selected codes only)

ASCII code	Character
65	A
66	B
67	C
68	D

(b) Some bit manipulation instructions are shown in the table:

Instruction		Explanation
Opcode	Operand	
AND	#n	Bitwise AND operation of the contents of ACC with the operand
AND	<address>	Bitwise AND operation of the contents of ACC with the contents of <address>
XOR	#n	Bitwise XOR operation of the contents of ACC with the operand
XOR	<address>	Bitwise XOR operation of the contents of ACC with the contents of <address>
OR	#n	Bitwise OR operation of the contents of ACC with the operand
OR	<address>	Bitwise OR operation of the contents of ACC with the contents of <address>
<address> can be an absolute address or a symbolic address # denotes a denary number, e.g. #123		

The contents of the memory address 300 are shown:

Bit Number	7	6	5	4	3	2	1	0
300	0	1	1	0	0	1	1	0

(i) The contents of memory address 300 represent an unsigned binary integer.

Write the denary value of the unsigned binary integer in memory address 300.

[1]

60. M/J 21/P12/Q6/a,b,c

6 A computer uses the ASCII character set.

(a) State the number of characters that can be represented by the ASCII character set and the extended ASCII character set.

ASCII

Extended ASCII

[2]

(b) Explain how a word such as 'HOUSE' is represented by the ASCII character set.

[2]

(c) Unicode is a different character set.

The Unicode value for the character '1' is denary value 49.

(i) Write the hexadecimal value for the Unicode character '1'.

[1]

(ii) Write the denary value for the Unicode character '5'.

[1]

61. M/J 21/P13/Q1/a,b,c

1 Anya scans an image into her computer for a school project.

(a) The scanned image is a bitmapped image.

(i) Complete the following table to describe the two terms about graphics.

Term	Description
Pixel	
File header	

[2]

(ii) The image is scanned with an image resolution of 1024×512 pixels, and a colour depth of 8 bits per pixel.

Calculate an estimate for the file size, giving your answer in mebibytes. Show your working.

Working

Answer mebibytes

[3]

(b) The image is compressed using lossless compression.

Identify **one** method of lossless compression that can be used to compress the image **and** describe how the method will reduce the file size.

Lossless compression method

Description

[3]

(c) One of the colours used in the image has the hexadecimal colour code:

#FC238A

FC is the amount of red, 23 is the amount of green and 8A is the amount of blue in the colour.

(i) Convert the hexadecimal code FC into denary.

[1]

(ii) The amount of green in binary is 00100011. This has the denary number 15 added to it to create a second colour.

Add the denary number 15 to the binary number 00100011 and give your answer in binary.

Perform the addition in binary. Show your working.

Working

Answer (in binary)

[3]

- (iii) Hexadecimal 23 in two's complement representation is 00100011. The denary number 10 needs to be subtracted from this value.

Subtract the denary number 10 from the two's complement representation 00100011.

Give your answer in binary. Show your working.

Working

Answer (in binary)

[3]

62. 9608 M/J 21/P11/Q2/a,b,c

- 2 Zak designs a logo for his company. He uses vector graphics software to create the logo.



- (a) One of the drawing objects in the logo is a circle.

Identify **four** properties of the circle.

- 1
- 2
- 3
- 4

[4]

- (b) Describe what is meant by a **drawing list** using the logo as an example.

[2]

- (c) Zak could have used a bitmapped image for the logo.

Describe **two** drawbacks of using a bitmapped image for the logo instead of a vector graphic.

Drawback 1

Drawback 2

[4]

63. 9608 M/J 21/P11/Q3/c

- 3 The table shows part of the instruction set for a processor. The processor has one general purpose register, the Accumulator (ACC), and an Index Register (IX).

Instruction		Explanation
Op code	Operand	
LDD	<address>	Direct addressing. Load the content of the location at the given address to ACC.
LDI	<address>	Indirect addressing. The address to be used is at the given address. Load the contents of this second address to ACC.
DEC	<register>	Subtract 1 from the contents of the register (ACC or IX).
CMP	<address>	Compare the contents of ACC with the contents of <address>.
JMP	<address>	Jump to the given address.
JPE	<address>	Following a compare instruction, jump to <address> if the compare was True.
STO	<address>	Store the contents of ACC at the given address.
END		Return control to the operating system.

The current contents of the main memory are:

Address	Instruction
100	LDD 200
101	CMP 201
102	JPE 106
103	DEC ACC
104	STO 200
105	JMP 101
106	END
...	
200	2
201	0
202	200

(c) Each instruction in the assembly language program is encoded in 16 bits (8-bit op code followed by an 8-bit operand).

(i) The instruction `CMP 201` has the operand 201.

Convert the operand 201 into 8-bit binary.

--	--	--	--	--	--	--	--

[1]

(ii) State the **maximum** number of op codes that can be represented using eight bits.

[1]

64. 9608 M/J 21/P11/Q8/b

8 Jay is developing a computer game that allows users to create stories.

(b) The game generates a story which is stored as a text file.

Jay compresses the text file using lossless compression before sending it by email to his friend.

(i) Identify **two** reasons for compressing the text file.

1

2

[2]

(ii) Explain the reasons why Jay compresses the text file with lossless compression instead of lossy compression.

[2]

65. 9608 M/J 21/P12/Q3/c

3 The table shows part of the instruction set for a processor. The processor has one general purpose register, the Accumulator (ACC), and an Index Register (IX).

Instruction		Explanation
Op code	Operand	
LDD	<address>	Direct addressing. Load the contents of the location at the given address to ACC.
LDI	<address>	Indirect addressing. The address to be used is at the given address. Load the contents of this second address to ACC.
STO	<address>	Store the contents of ACC at the given address.
ADD	<address>	Add the contents of the given address to ACC.
INC	<register>	Add 1 to the contents of the register (ACC or IX).
JMP	<address>	Jump to the given address.
CMP	<address>	Compare the contents of ACC with the contents of <address>.
JPE	<address>	Following a compare instruction, jump to <address> if the compare was True.
END		Return control to the operating system.

The current contents of the main memory are:

Address	Instruction
50	LDI 103
51	CMP 101
52	JPE 59
53	ADD 102
54	STO 102
55	LDD 100
56	INC ACC
57	STO 100
58	JMP 51
59	ADD 102
60	STO 102
61	END
...	
100	1
101	3
102	0
103	100

- (c) Each instruction in the assembly language program is encoded in 16 bits (8-bit op code followed by an 8-bit operand).

- (i) The instruction `JPE 59` has the operand 59.

Convert the operand 59 into 8-bit binary.

--	--	--	--	--	--	--	--

[1]

- (ii) Convert the denary value 59 into hexadecimal.

[1]

66. 9608 M/J 21/P12/Q7/b,c

- 7 Sam is a photographer. She has an image library of over 10 000 images. She stores the images on a high capacity magnetic hard disk.

(b) The images are stored as bitmap files.

Identify **four** items that will be stored in the header of a bitmap file.

1

2

3

4

[4]

(c) The bitmap images are compressed for use on a website.

Tick (✓) **one** box to select the **most appropriate** type of compression for the images used on the website and justify your answer.

Lossy	
Lossless	

Justification

[3]

67. 9608 M/J 21/P13/Q3/c

- 3 The following table shows part of the instruction set for a processor. The processor has one general purpose register, the Accumulator (ACC), and an Index Register (IX).

Instruction		Explanation
Op code	Operand	
LDX	<address>	Indexed addressing. Form the address from <address> + the contents of the index register. Copy the contents of this calculated address to ACC.
LDR	#n	Immediate addressing. Load the number n to IX.
DEC	<register>	Subtract 1 from the contents of the register (ACC or IX).
JMP	<address>	Jump to the given address.
CMP	<address>	Compare the contents of ACC with the contents of <address>.
JPE	<address>	Following a compare instruction, jump to <address> if the compare was True.
OUT		Output to the screen the character whose ASCII value is stored in ACC.
END		Return control to the operating system.

The current contents of the main memory and selected values from the ASCII character set are:

Address Instruction

75	LDR #2
76	LDX 180
77	CMP #0
78	JPE 82
79	OUT
80	DEC IX
81	JMP 76
82	END
...	
180	0
181	41
182	71
183	40
184	70
185	43
186	69

ASCII code table (selected codes only)

ASCII code	Character
36	\$
40	(
41	)
43	+
69	E
70	F
71	G

- (c) Each instruction in the assembly language program is encoded in 16 bits (8-bit op code followed by an 8-bit operand).

- (i) The instruction `LDX 234` has the operand 234.

Convert the operand 234 into 8-bit binary.

--	--	--	--	--	--	--	--

[1]

- (ii) Convert the denary value 234 into hexadecimal.

[1]

- (iii) The contents of memory address 190 represent a two's complement binary integer.

Address

190	1	1	0	1	1	0	0	1
-----	---	---	---	---	---	---	---	---

Convert the value in memory address 190 into denary.

[1]

68. 9608 M/J 21/P13/Q9/a

9 A website streams music and videos.

- (a) Two descriptions about how sound is encoded and represented on a computer are given.

Give the correct term for each description.

- (i) "The number of samples taken per unit time"

Term

[1]

- (ii) "The number of bits used to encode each sample"

Term

[1]

69. 9608 M/J 21/P21/Q1/a

- 1 (a) (i) State how characters are represented using the ASCII character set.

[2]

- (ii) String values may be represented by a sequence of ASCII characters.

The following table shows consecutive memory locations.

Complete the table by adding the values to show how the string "FADED" may be stored in memory using the ASCII character set.

Refer to the **Appendix** on pages 22–23 for the list of built-in pseudocode functions and operators, which includes a reference to the `ASC ()` function.

Memory location	ASCII character value
100	
101	
102	
103	
104	

[2]

Appendix of this paper:

Appendix

Built-in functions (pseudocode)

Each function returns an error if the function call is not properly formed.

`MID(ThisString : STRING, x : INTEGER, y : INTEGER)` RETURNS STRING
returns a string of length `y` starting at position `x` from `ThisString`

Example: `MID("ABCDEFGH", 2, 3)` returns "BCD"

LENGTH(ThisString : STRING) RETURNS INTEGER
returns the integer value representing the length of ThisString

Example: LENGTH("Happy Days") returns 10

LEFT(ThisString : STRING, x : INTEGER) RETURNS STRING
returns leftmost x characters from ThisString

Example: LEFT("ABCDEFGH", 3) returns "ABC"

RIGHT(ThisString : STRING, x : INTEGER) RETURNS STRING
returns rightmost x characters from ThisString

Example: RIGHT("ABCDEFGH", 3) returns "FGH"

INT(x : REAL) RETURNS INTEGER
returns the integer part of x

Example: INT(27.5415) returns 27

LCASE(ThisChar : CHAR) RETURNS CHAR
returns the character value representing the lower case equivalent of ThisChar
If ThisChar is not an upper case alphabetic character, it is returned unchanged.

Example: LCASE('W') returns 'w'

ASC(ThisChar : CHAR) RETURNS INTEGER
returns the ASCII value of character ThisChar

Example: ASC('A') returns 65

RAND(x : INTEGER) RETURNS REAL
returns a real number in the range 0 to x (not inclusive of x).

Example: RAND(87) could return 35.43

NUM_TO_STRING(x : REAL) RETURNS STRING
returns a string representation of a numeric value.
Note: This function will also work if x is of type INTEGER

Example: NUM_TO_STRING(87.5) returns "87.5"

Operators (pseudocode)

Operator	Description
&	Concatenates (joins) two strings Example: "Summer" & " " & "Pudding" produces "Summer Pudding"
AND	Performs a logical AND on two Boolean values Example: TRUE AND FALSE produces FALSE
OR	Performs a logical OR on two Boolean values Example: TRUE OR FALSE produces TRUE

70. 9608 M/J 21/P22/Q1/b

- (b) State why characters need to be represented in ASCII or Unicode before they can be processed.

[1]

71. M/J 20/P11/Q4/a,b,c

- 4 A digital camera takes a bitmap image. The image is 2000 pixels wide by 1000 pixels high with a colour depth of 24-bits.

- (a) Calculate an estimate of the file size for the image. Give your answer in megabytes. Show your working.

Working

Answer MB

[3]

- (b) A second image is taken, this time in black and white. It has the same number of pixels, but the file size is smaller.

Explain why the file size is smaller.

[2]

- (c) The digital camera allows a user to add text to an image. The text is encoded as ASCII values.

The table shows the ASCII denary values for five characters.

Character	ASCII denary value
a	97
b	98
c	99
d	100
e	101

- (i) Give the 8-bit binary value for the ASCII character 'b'.

[1]

- (ii) Complete the table by writing the ASCII denary value for the character 't' and its hexadecimal equivalent.

Character	t
ASCII denary value	
Hexadecimal value	

[2]

72. M/J 20/P11/Q9/b

9 Utility programs are examples of system software.

(b) File compression is one example of a utility program.

Tick (✓) **one** box on each row to indicate whether the action is an example of lossy compression or lossless compression.

Action	Lossy	Lossless
Reducing the resolution of an image		
Using run-length encoding on a text file		
Reducing the sampling rate of a sound file		

[1]

73. M/J 20/P12/Q1/b

1 Samira is creating an interactive, multimedia presentation for the entrance to her hotel.

(b) Samira uses a computer to draw a logo for her hotel and saves it as a vector graphic. The logo will be placed on the multimedia presentation and elsewhere, such as on signs at the entrance of the hotel.

Samira emails the logo to a company that prints signs, and other documentation for the hotel.

(i) Describe how the logo is represented by the computer.

[3]

(ii) State **two** reasons why the hotel **logo** is saved as a vector graphic instead of a bitmapped graphic.

1

2

[2]

74. M/J 20/P12/Q2/a,b

2 Amir has created a sound file using his desktop computer.

(a) Complete the table by writing the missing definitions and term about sound.

Term	Definition
Sampling	
.....	The number of samples per unit time
Sampling resolution	

[3]

(b) The file is too large to be emailed and the file size needs to be reduced.

(i) Name **one** lossless compression technique that can be used to reduce the size of the sound file.

[1]

(ii) Describe **one** lossy compression technique that can be used to reduce the size of the sound file.

[2]

75. M/J 20/P12/Q5/b

- (b) The assembler translates assembly code into machine code.

The table shows the denary values for three assembler op codes.

Op code	Denary value
LDD	194
ADD	200
STO	205

- (i) Convert the denary value for the op code LDD into 8-bit binary.

--	--	--	--	--	--	--	--

[1]

- (ii) Convert the denary value for the op code STO into hexadecimal.

[1]

- (iii) State why the denary value for the op code ADD cannot be represented in 8-bit two's complement form. Justify your answer.

[2]

76. M/J 20/P13/Q5/b

- 5 Wei is developing a program.

- (b) Wei wants to compress the source code to transport it to another computer.

Identify the **most appropriate** compression technique he should use.

Justify your choice.

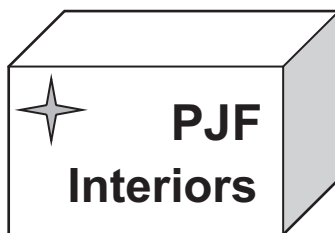
Compression technique

Justification

[3]

77. M/J 20/P13/Q7/b,c,d

7 Xiaoming created the following logo using bitmapped graphics software.



- (b) The finished logo is 160 pixels wide and 160 pixels high. The image has a colour depth of 3 bytes per pixel.

Calculate an estimate of the file size for the logo. Give your answer in kilobytes. Show your working.

Working

Answer KB

[3]

- (c) Xiaoming needs to use his logo on his business card, on his website and on large display boards. He is told that he should have created a vector graphic logo instead of a bitmapped graphic logo.

Describe **one** benefit of creating a vector graphic logo instead of a bitmapped graphic logo.

[2]

- (d) The hexadecimal colour value of the background of Xiaoming's website is:

913C8E

Complete the following table by converting each hexadecimal value to denary value.

	Red	Green	Blue
Hexadecimal value	91	3C	8E
Denary value			

[2]

78. O/N 19/P11/Q5/a,b,c,d,e

- 5 (a) The bit depth of an image dictates how many different colours can be represented by each pixel.

(i) State the number of different colours that can be represented by a bit depth of 8 bits.

[1]

(ii) One binary colour is represented by 0100 1110

Convert the unsigned binary number 0100 1110 into denary.

[1]

- (b) Convert the denary number –194 into 12-bit two’s complement.

[1]

- (c) (i) Convert the Binary Coded Decimal (BCD) value 0110 1001 into denary.

[1]

- (ii) Identify **one** practical application where BCD is used.

[1]

- (d) One example of a character set used by computers is ASCII.

Describe how one character is represented in a character set.

[2]

- (e) Data can be compressed using either lossy or lossless compression.

Tick (✓) **one** box in each scenario to identify whether lossy or lossless compression should be used. Justify your choice.

- (i) A program written in a high-level language.

Lossy	Lossless

Justification

[2]

- (ii) A photograph that needs to be emailed to a friend.

Lossy	Lossless

Justification

[2]

- (iii) You need to upload a video that you have created to a website.

Lossy	Lossless

Justification

[2]

79. O/N 19/P12/Q6/d(i)

- 6 Dominic uses a tablet computer to complete work. He records videos of his work to send to his colleagues to watch at a later date.

(d) Dominic's tablet captures a video of Dominic to send to other people. The video is made of a sequence of images and a sound file.

(i) Describe how the images and sound are encoded into a digital form.

Images

Sound

[4]

80. O/N 19/P13/Q2/b(i,ii)

- 2 Leonardo's mobile phone has an operating system (OS).

(b) Leonardo uses the mobile phone to record his voice.

(i) Describe how sound sampling is used by the mobile phone to encode the sound.

[2]

(ii) Leonardo records his voice twice. Each recording is the same length and has the same sampling resolution.

The first recording has a sampling rate of 44 100 Hz. The second recording has a sampling rate of 21 000 Hz.

Describe how the different sampling rates will affect the recording and the sound file.

[2]

81. O/N 19/P13/Q4/a,b,c,d,e

- 4 (a) Convert the unsigned binary number 0101 1111 1100 into denary.

[1]

(b) Convert the denary number -239 into 12-bit two's complement.

[1]

(c) Convert the two's complement number 0110 0101 into denary.

[1]

(d) Convert the Binary Coded Decimal (BCD) value 0110 0101 into denary.

[1]

(e) Convert the denary number 222 into hexadecimal.

[1]

82. M/J 19/P11/Q1/d

1 Devices connected to the Internet have IP (Internet Protocol) addresses.

(d) A recording of a concert is stored as a file. The file is compressed using lossy compression before it is streamed to users.

(i) State why this file needs to be compressed.

[1]

(ii) Define the term **lossy compression**.

[1]

(iii) The file could be compressed using lossless compression.

Explain why lossy compression is a more appropriate compression technique than lossless for this file.

[3]

83. M/J 19/P11/Q6/d

6 A student records a video using a digital camera.

(d) A sound track is recorded for the video.

(i) Describe how a computer encodes the sound track.

[3]

(ii) Explain how the sampling rate and sampling resolution affect the file size of the sound track.

Sampling rate

Sampling resolution

[2]

84. M/J 19/P12/Q3/d

3 The fetch-execute cycle is shown in register transfer notation.

01 $\text{MAR} \leftarrow [\text{PC}]$

02 $\text{PC} \leftarrow [\text{PC}] - 1$

03 $\text{MDR} \leftarrow [\text{MAR}]$

04 $\text{CIR} \leftarrow [\text{MAR}]$

(d) The ASCII character code for 'A' is 65 in denary.

(i) Convert the denary ASCII character code for 'A' into 8-bit binary.

--	--	--	--	--	--	--	--

[1]

(ii) Convert the denary ASCII character code for 'A' into hexadecimal.

[1]

(iii) The Unicode character code for 'G' is 0047 in hexadecimal.

State, in hexadecimal, the Unicode character code for 'D'.

[1]

85. M/J 19/P13/Q5/a,c

5 Xander creates a presentation that includes images, video and sound.

- (a)** The images are bitmap images. A bitmap image can be made up of any number of colours. Each colour is represented by a unique binary number.

Draw **one** line from **each** box on the left, to the correct box on the right to identify the minimum number of bits needed to store each maximum number of colours.

Maximum number of colours

68

256

127

2

249

Minimum number of bits

1

2

3

7

8

9

[3]

- (c)** The sound track has a sampling rate of 88.2 kHz and a sampling resolution of 32 bits.

State what is meant by a **sampling rate of 88.2 kHz** and a **sampling resolution of 32 bits**.

Sampling rate of 88.2 kHz

Sampling resolution of 32 bits

[2]

86. O/N 18/P11/Q1/b

- 1 A student is creating a short video and needs to record music to play in the background.

(b) An analogue-to-digital converter uses sampling to encode the sound.

Explain how different sampling resolutions affect the sound file and the sound it represents.

[3]

87. O/N 18/P11/Q4/c

- 4 The table shows assembly language instructions for a processor which has one general purpose register, the Accumulator (ACC) and an index register (IX).

Instruction		Explanation
Op code	Operand	
LDD	<address>	Direct addressing. Load the contents of the location at the given address to ACC.
LDX	<address>	Indexed addressing. Form the address from <address> + the contents of the Index Register. Copy the contents of this calculated address to ACC.
LDR	#n	Immediate addressing. Load the number n to IX.
STO	<address>	Store contents of ACC at the given address.
ADD	<address>	Add the contents of the given address to ACC.
INC	<register>	Add 1 to the contents of the register (ACC or IX).
DEC	<register>	Subtract 1 from the contents of the register (ACC or IX).
CMP	<address>	Compare contents of ACC with contents of <address>.
JPE	<address>	Following a compare instruction, jump to <address> if the compare was True.
JPN	<address>	Following a compare instruction, jump to <address> if the compare was False.
JMP	<address>	Jump to the given address.
OUT		Output to the screen the character whose ASCII value is stored in ACC.
END		Return control to the operating system.

(c) The current contents of a general purpose register (X) are:

X	1	0	1	1	1	0	1	0
---	---	---	---	---	---	---	---	---

(i) The contents of X represent an unsigned binary integer.

Convert the value in X into denary.

[1]

(ii) The contents of X represent an unsigned binary integer.

Convert the value in X into hexadecimal.

[1]

- (iii) The contents of X represent a two's complement binary integer.

Convert the value in X into denary.

[1]

88. O/N 18/P12/Q1/a,b,c

- 1 A company is designing a website.

- (a) The company creates a 4-colour bitmap image for the website as shown.

Each colour is represented by a letter, for example, G = grey, K = black.

G	R	G	K	W	R
G	R	G	K	W	R
G	R	G	K	W	R
G	R	G	K	W	R
G	G	G	K	K	R
W	W	W	W	K	R

- (i) State the minimum number of bits needed to represent each pixel in the image in **part (a)**.

[1]

- (ii) Calculate the minimum file size of the image shown in **part (a)**. Show your working.

Working

File size

[3]

- (b) The company takes a photograph of their office to put on the website. The photograph has a resolution of 1000 pixels by 1000 pixels. Two bytes per pixel are used to represent the colours.

- (i) Estimate the file size of the photograph in megabytes. Show your working.

Working

Estimated file size

[4]

- (ii) The file size of the photograph needs to be reduced before it is placed on the website.

Draw lines to link each method of reducing the file size of the image to:

- its description and
- its compression type, where appropriate.

Description	Method	Compression type
Removes pixels	Crop the photograph	Lossy
Reduces number of pixels per inch	Use run-length encoding	Lossless
Uses fewer bits per pixel	Use fewer colours	
Stores colour code and count of repetitions		

[5]

(c) The company has created a logo for the website. The logo is a vector graphic.

Describe **two** reasons why a vector graphic is a sensible choice for the logo.

Reason 1

Reason 2

[4]

89. O/N 18/P12/Q3/c

- 3** The following table shows assembly language instructions for a processor which has one general purpose register, the Accumulator (ACC) and an Index Register (IX).

Instruction		Explanation
Op code	Operand	
LDD	<address>	Direct addressing. Load the contents of the location at the given address to ACC.
LDX	<address>	Indexed addressing. Form the address from <address> + the contents of the Index Register. Copy the contents of this calculated address to ACC.
LDR	#n	Immediate addressing. Load the number n to IX.
STO	<address>	Store contents of ACC at the given address.
ADD	<address>	Add the contents of the given address to ACC.
INC	<register>	Add 1 to the contents of the register (ACC or IX).
DEC	<register>	Subtract 1 from the contents of the register (ACC or IX).
CMP	<address>	Compare contents of ACC with contents of <address>.
JPE	<address>	Following compare instruction, jump to <address> if the compare was True.
JPN	<address>	Following compare instruction, jump to <address> if the compare was False.
JMP	<address>	Jump to the given address.
OUT		Output to the screen the character whose ASCII value is stored in ACC.
END		Return control to the operating system.

- (c)** The current contents of a general purpose register (X) are:

X	1	1	0	0	0	0	0	1
---	---	---	---	---	---	---	---	---

- (i)** The contents of X represent an unsigned binary integer.

Convert the value in X into denary.

[1]

- (ii)** The contents of X represent an unsigned binary integer.

Convert the value in X into hexadecimal.

[1]

- (iii)** The contents of X represent a two's complement binary integer.

Convert the value in X into denary.

[1]

90. O/N 18/P13/Q1/a,b,c

- 1** A product designer is creating a poster.

- (a)** The designer creates a 6-colour bitmap image for the poster as shown.

Each colour is represented by a letter, for example, R = red, B = blue.

R	R	P	P	P	G
B	R	R	P	G	G
B	W	B	B	O	O
B	W	W	P	P	O
B	B	R	P	G	O
B	R	R	P	G	O

- (i) State the minimum number of bits needed to represent each pixel in the image in **part (a)**.

[1]

- (ii) Calculate the minimum file size of the image shown in **part (a)**. Show your working.

Working

File size

[3]

- (b) (i) The designer takes a photograph to put on the poster. The photograph has a resolution of 50 000 pixels by 50 000 pixels. The colours are represented using 4 bytes per pixel.

Estimate the file size of the photograph in gigabytes. Show your working.

Working

Estimated file size

[4]

- (ii) The photograph needs to be sent by email but the file size is too big. It needs to be compressed.

The table lists several methods of making an image file size smaller.

Tick (✓) **one** box on each row to indicate whether each method is lossy or lossless.

Compression method	Lossy	Lossless
Cropping the image		
Reducing the resolution of the image		
Using run-length encoding (RLE)		
Reducing the colour depth of the image		

[4]

- (c) Explain how run-length encoding would compress the image in **part (a)**.

[3]

91. O/N 18/P13/Q2/b(i,ii,iii)

- 2 The following table shows assembly language instructions for a processor which has one general purpose register, the Accumulator (ACC) and an Index Register (IX).

Instruction		Explanation
Op code	Operand	
LDD	<address>	Direct addressing. Load the contents of the location at the given address to ACC.
LDX	<address>	Indexed addressing. Form the address from <address> + the contents of the Index Register. Copy the contents of this calculated address to ACC.
LDR	#n	Immediate addressing. Load the number n to IX.
STO	<address>	Store contents of ACC at the given address.
ADD	<address>	Add the contents of the given address to ACC.
INC	<register>	Add 1 to the contents of the register (ACC or IX).
DEC	<register>	Subtract 1 from the contents of the register (ACC or IX).
CMP	<address>	Compare contents of ACC with contents of <address>.
JPE	<address>	Following compare instruction, jump to <address> if the compare was True.
JPN	<address>	Following compare instruction, jump to <address> if the compare was False.
JMP	<address>	Jump to the given address.
OUT		Output to the screen the character whose ASCII value is stored in ACC.
END		Return control to the operating system.

- (b) The current contents of a general purpose register (X) are:

X	1	1	1	1	0	0	1	0
---	---	---	---	---	---	---	---	---

- (i) The contents of X represent an unsigned binary integer.

Convert the value in X into denary.

[1]

- (ii) The contents of X represent an unsigned binary integer.

Convert the value in X into hexadecimal.

[1]

- (iii) The contents of X represent a two's complement binary integer.

Convert the value in X into denary.

[1]

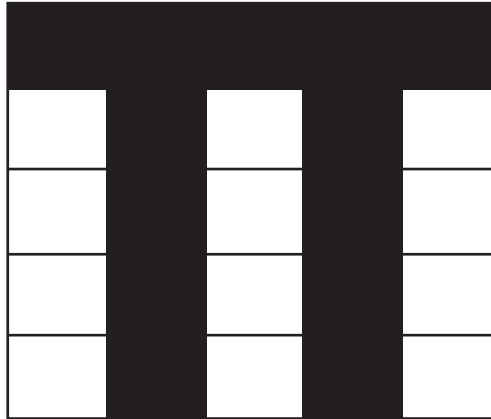
92. M/J 18/P11/Q2/a,b,c,d

2 A logo is designed as a bitmap image.

(a) Describe what is meant by a **bitmap image**.

[2]

(b) A black and white bitmap image is shown.



(i) Explain how a computer can store this bitmap image.

[2]

(ii) The image is compressed before it is attached to an email.

Explain how run-length encoding (RLE) will compress the image.

[2]

(c) The finished logo is 500 pixels by 1000 pixels and uses 35 different colours.

Estimate the file size for the logo. Give your answer in kilobytes. Show your working.

Working

Answer

[4]

(d) The logo is redesigned as a vector graphic.

State **two** benefits of a vector graphic compared to a bitmap image. Give a reason for each benefit.

Benefit 1

Reason 1

Benefit 2

Reason 2

[4]

93. M/J 18/P11/Q8/c

8 The Von Neumann model uses a series of registers.

(c) X is a register. The current contents of X are:

1	0	0	0	0	1	1	1
---	---	---	---	---	---	---	---

(i) The current contents of register X represent an unsigned binary integer.

Convert the value in X into denary.

[1]

(ii) The current contents of register X represent a Binary Coded Decimal.

Convert the value in X into denary.

[1]

(iii) The current contents of register X stores a two's complement binary integer.

Convert the value in X into denary.

[1]

94. M/J 18/P12/Q4/a,b

- 4 (a) The Accumulator is a register. The current contents of the Accumulator are:

1	1	0	1	1	0	1	1
---	---	---	---	---	---	---	---

The current contents of the Accumulator represent an unsigned binary integer.

- (i) Convert the value in the Accumulator into denary.

[1]

- (ii) Convert the value in the Accumulator into hexadecimal.

[1]

- (iii) The current contents of the Accumulator represent a two's complement binary integer.

Convert the value in the Accumulator into denary.

[1]

- (b) The binary integer represents a character from the computer's character set.

- (i) Define the term **character set**.

[1]

- (ii) Explain the differences between the **ASCII** and **Unicode** character sets.

[2]

- (iii) The ASCII code for 'A' is 41 in hexadecimal.

Calculate the ASCII code in hexadecimal for 'Z'. Show your working.

Working

ASCII code in hexadecimal for 'Z'

[2]

95. M/J 18/P12/Q5/a,b,c

5 A student has recorded a sound track for a short film.

(a) Explain how an analogue sound wave is sampled to convert it into digital format.

[3]

(b) Explain the effects of increasing the sampling resolution on the sound file.

[2]

(c) The original sound was sampled at 44.1 kHz. The sample rate is changed to 22.05 kHz.

Explain the effects of this change on the sound file.

[3]

96. M/J 18/P13/Q3/c

3 A computer is designed using the Von Neumann model.

(c) H is a register. The current contents of H are:

1	1	0	0	0	0	0	1
---	---	---	---	---	---	---	---

The current contents of register H represent an unsigned binary integer.

(i) Convert the value in register H into denary.

[1]

(ii) Convert the value in register H into hexadecimal.

[1]

(iii) The current contents of register H represent a two's complement binary integer.

Convert the value in register H into denary.

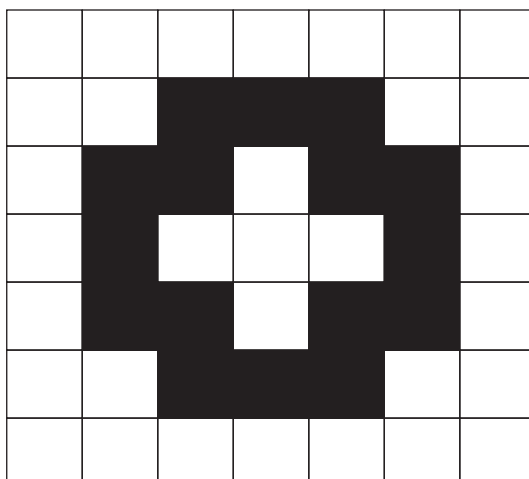
[1]

(iv) State why register H does not currently contain a Binary Coded Decimal (BCD).

[1]

97. M/J 18/P13/Q6/a,b,c,d

6 A black and white bitmap image is shown.



(a) State the **minimum** number of bits needed to represent each pixel in this image.

[1]

(b) Run-length encoding (RLE) is used to store the image with the following colour codes.

Colour	Code
Black	1A
White	3B

Show how run-length encoding is used to store the image.

[3]

(c) An image has 30 different colours.

State the **minimum** number of bits needed to represent each pixel in the 30-colour image.

[1]

(d) When the image is saved, a header is added to the file.

State the purpose of the **file header**. Give **two** examples of the file header contents.

Purpose

Example 1

Example 2

[3]

98. O/N 17/P13/Q1/a,b

- 1 (a)** Each of the following bytes represents an integer in two's complement form.

State the denary value.

(i) 0111 0111 Denary [1]

(ii) 1000 1000 Denary [1]

- (iii)** Express the following integer in two's complement form.

-17

--	--	--	--	--	--	--	--

[1]

- (iv)** State in denary, the range of integer values that it is possible to represent in two's complement integers using a single byte.

Lowest value

Highest value [1]

- (b) (i)** Convert the following denary integer into Binary Coded Decimal (BCD).

653

[1]

- (ii)** A 3-digit BCD representation has been incorrectly copied. It is shown as:

0	1	0	0	1	1	1	0	0	0	1	0
---	---	---	---	---	---	---	---	---	---	---	---

State how you can recognise that this is not a valid BCD representation.

[1]

- (iii)** Describe a practical application where BCD is used.

[1]

99. O/N 17/P22/Q1/b(i)

- (b) (i) In the ASCII character set 'A' is represented by the value 65. The values representing the other characters of the alphabet follow in sequence, so 'B' is represented by 66, 'C' by 67 and so on.

The following table represents consecutive memory locations. Each memory location stores one byte.

Complete the table to show how the string "CAGE" may be stored in memory using the ASCII set.

Address	Data
100	
101	
102	
103	
104	
105	

[2]

100. M/J 17/P11/Q3/a,b

- 3 (a)** A computer has a microphone and captures a voice recording using sound recording software.

Before making a recording, the user can select the sampling rate.

Define the term **sampling rate**. Explain how the sampling rate will influence the accuracy of the digitised sound.

Sampling rate

Explanation

[2]

- (b)** The computer also has bitmap software.

- (i)** Define the terms **pixel** and **screen resolution**.

Pixel

Screen resolution

[2]

- (ii)** A picture has been drawn and is saved as a monochrome bitmap image.

State how many pixels are stored in one byte.

[1]

- (iii)** A second picture has width 2048 pixels and height 512 pixels. It is saved as a 256-colour image.

Calculate the file size in kilobytes.

Show your working.

[3]

- (iv)** The actual bitmap file size will be larger than your calculated value.

State another data item that the bitmap file stores in addition to the pixel data.

[1]

101. M/J 17/P11/Q4/d(i)

- 4 The following table shows part of the instruction set for a processor. The processor has one general purpose register, the Accumulator (ACC) and an Index Register (IX).

Instruction		Op code (binary)	Explanation
Op code (mnemonic)	Operand		
LDM	#n	0000 0001	Immediate addressing. Load the denary number <i>n</i> to ACC.
LDD	<address>	0000 0010	Direct addressing. Load the contents of the location at the given address to ACC.
LDI	<address>	0000 0101	Indirect addressing. At the given address is the address to be used. Load the contents of this second address to ACC.
LDX	<address>	0000 0110	Indexed addressing. Form the address from <address> + the contents of the Index Register (IX). Copy the contents of this calculated address to ACC.
LDR	#n	0000 0111	Immediate addressing. Load number <i>n</i> to IX.
STO	<address>	0000 1111	Store the contents of ACC at the given address.

The following diagram shows the contents of a section of main memory and the Index Register (IX).

(d) Computer scientists often write binary representations in hexadecimal.

- (i) Write the hexadecimal representation for this instruction:

0	0	0	0	0	1	1	1	1	1	0	0	0	0	1	0
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

[2]

102. M/J 17/P12/Q3/a,b

- 3 (a)** A computer has a microphone and captures a voice recording using sound editing software.

The user can select the sampling resolution before making a recording.

Define the term **sampling resolution**. Explain how the sampling resolution will affect the accuracy of the digitised sound.

Sampling resolution

Explanation

[3]

- (b)** The computer also has bitmap software.

- (i)** Define the term **image resolution**.

[1]

- (ii)** A picture is drawn and is saved as a 16-colour bitmap image.

State how many bits are used to encode the data for one pixel.

[1]

- (iii)** A second picture has width 8192 pixels and height 256 pixels. It is saved as a 256-colour bitmap.

Calculate the file size in kilobytes.

Show your working.

[3]

- (iv)** The actual bitmap file size will be larger than your calculated value as a bitmap file has a file header.

State **two** items of data that are stored in the file header.

1

2

[2]

103. M/J 17/P12/Q5/c(i)

- 5 The following table shows part of the instruction set for a processor. The processor has one general purpose register, the Accumulator (ACC), and an Index Register (IX).

Instruction			Explanation
Op code (mnemonic)	Operand	Op code (binary)	
LDD <address>		0001 0011	Direct addressing. Load the contents of the location at the given address to the Accumulator (ACC).
LDI <address>		0001 0100	Indirect addressing. The address to be used is at the given address. Load the contents of this second address to ACC.
LDX <address>		0001 0101	Indexed addressing. Form the address from <address> + the contents of the Index Register. Copy the contents of this calculated address to ACC.
LDM #n		0001 0010	Immediate addressing. Load the denary number n to ACC.
LDR #n		0001 0110	Immediate addressing. Load denary number n to the Index Register (IX).
STO <address>		0000 0111	Store the contents of ACC at the given address.

The following diagram shows the contents of a section of main memory and the Index Register (IX).

- (c) Computer scientists often write binary representations in hexadecimal.

- (i) Write the hexadecimal representation for the following instruction.

0	0	0	1	0	1	0	0	0	1	0	1	1	1	1	0
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

[2]

104. M/J 17/P13/Q3/a,b

- 3 (a)** A computer has a microphone and captures a voice recording using sound recording software.

Before making a recording, the user can select the sampling rate.

Define the term **sampling rate**. Explain how the sampling rate will influence the accuracy of the digitised sound.

Sampling rate

Explanation

[2]

- (b)** The computer also has bitmap software.

- (i)** Define the terms **pixel** and **screen resolution**.

Pixel

Screen resolution

[2]

- (ii)** A picture has been drawn and is saved as a monochrome bitmap image.

State how many pixels are stored in one byte.

[1]

- (iii)** A second picture has width 2048 pixels and height 512 pixels. It is saved as a 256-colour image.

Calculate the file size in kilobytes.

Show your working.

[3]

- (iv)** The actual bitmap file size will be larger than your calculated value.

State another data item that the bitmap file stores in addition to the pixel data.

[1]

105. M/J 17/P13/Q4/d(i)

- 4 The following table shows part of the instruction set for a processor. The processor has one general purpose register, the Accumulator (ACC) and an Index Register (IX).

Instruction		Op code (binary)	Explanation
Op code (mnemonic)	Operand		
LDM	#n	0000 0001	Immediate addressing. Load the denary number <i>n</i> to ACC.
LDD	<address>	0000 0010	Direct addressing. Load the contents of the location at the given address to ACC.
LDI	<address>	0000 0101	Indirect addressing. At the given address is the address to be used. Load the contents of this second address to ACC.
LDX	<address>	0000 0110	Indexed addressing. Form the address from <address> + the contents of the Index Register (IX). Copy the contents of this calculated address to ACC.
LDR	#n	0000 0111	Immediate addressing. Load number <i>n</i> to IX.
STO	<address>	0000 1111	Store the contents of ACC at the given address.

The following diagram shows the contents of a section of main memory and the Index Register (IX).

(d) Computer scientists often write binary representations in hexadecimal.

- (i) Write the hexadecimal representation for this instruction:

0	0	0	0	0	1	1	1	1	1	0	0	0	0	1	0
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

[2]

106. O/N 16/P11/Q3/a,b

- 3 (a) (i)** Convert the denary number 46 to an 8-bit binary integer.

[1]

- (ii)** Convert the denary integer –46 to an 8-bit two's complement form.

[1]

- (iii)** Convert the denary number 46 into hexadecimal.

[1]

- (b)** Binary Coded Decimal (BCD) is another way of representing numbers.

- (i)** Describe how denary integers larger than 9 can be converted into BCD.
Give an example in your answer.

[2]

- (ii)** Describe how an 8-bit BCD representation can be converted into a denary integer.
Give an example in your answer.

[2]

107. O/N 16/P11/Q7/b

- 7** A small company produces scientific magazines. The owner buys some new desktop computers. The computers are used to store thousands of colour images (diagrams and photographs). All the computers have Internet access.

- (b)** The images contained in the magazines are produced using either bitmap or vector graphics software.

Give **four** differences between bitmap and vector graphics.

1

2

3

4

[4]

108. O/N 16/P11/Q8/c

- 8** The table shows assembly language instructions for a processor which has one general purpose register, the Accumulator (ACC) and an Index Register (IX).

Instruction		Explanation
Op code	Operand	
LDD	<address>	Direct addressing. Load the contents of the given address to ACC.
LDX	<address>	Indexed addressing. Form the address from <address> + the contents of the index register. Copy the contents of this calculated address to ACC.
STO	<address>	Store contents of ACC at the given address.
ADD	<address>	Add the contents of the given address to ACC.
CMP	<address>	Compare contents of ACC with contents of <address>
JPE	<address>	Following a compare instruction, jump to <address> if the compare was True.
JPN	<address>	Following a compare instruction, jump to <address> if the compare was False.
JMP	<address>	Jump to the given address.
OUT		Output to the screen the character whose ASCII value is stored in ACC.
END		Return control to the operating system.

The diagram shows the contents of the main memory:

Main memory	
800	0110 0100
801	0111 1100
802	1001 0111
803	0111 0011
804	1001 0000
805	0011 1111
806	0000 1110
807	1110 1000
808	1000 1110
809	1100 0010
:	
:	
2000	1011 0101

(c) The program used the ASCII coding system for character codes. An alternative coding system is Unicode.

(i) Give **two** disadvantages of using ASCII code.

1

2

[2]

(ii) Describe how Unicode is designed to overcome the disadvantages of ASCII.

[2]

109. O/N 16/P12/Q4

4 Hexadecimal, Binary Coded Decimal (BCD) and binary values are shown below.

Draw a line to link each value to its correct denary value.

Hexadecimal, BCD, binary**Denary**

Hexadecimal:

3A

BCD representation:

0100 1001

Binary integer:

01011101

Two's complement
binary integer:

11000001

93

-65

58

-63

73

49

-93

[4]

110. O/N 16/P13/Q3/a,b

- 3 (a) (i)** Convert the denary number 46 to an 8-bit binary integer.

[1]

- (ii)** Convert the denary integer –46 to an 8-bit two's complement form.

[1]

- (iii)** Convert the denary number 46 into hexadecimal.

[1]

- (b)** Binary Coded Decimal (BCD) is another way of representing numbers.

- (i)** Describe how denary integers larger than 9 can be converted into BCD.
Give an example in your answer.

[2]

- (ii)** Describe how an 8-bit BCD representation can be converted into a denary integer.
Give an example in your answer.

[2]

111. O/N 16/P13/Q7/b

- 7** A small company produces scientific magazines. The owner buys some new desktop computers. The computers are used to store thousands of colour images (diagrams and photographs). All the computers have Internet access.

- (b)** The images contained in the magazines are produced using either bitmap or vector graphics software.

Give **four** differences between bitmap and vector graphics.

1

2

3

4

[4]

112. O/N 16/P13/Q8/c

- 8** The table shows assembly language instructions for a processor which has one general purpose register, the Accumulator (ACC) and an Index Register (IX).

Instruction		Explanation
Op code	Operand	
LDD	<address>	Direct addressing. Load the contents of the given address to ACC.
LDX	<address>	Indexed addressing. Form the address from <address> + the contents of the index register. Copy the contents of this calculated address to ACC.
STO	<address>	Store contents of ACC at the given address.
ADD	<address>	Add the contents of the given address to ACC.
CMP	<address>	Compare contents of ACC with contents of <address>
JPE	<address>	Following a compare instruction, jump to <address> if the compare was True.
JPN	<address>	Following a compare instruction, jump to <address> if the compare was False.
JMP	<address>	Jump to the given address.
OUT		Output to the screen the character whose ASCII value is stored in ACC.
END		Return control to the operating system.

The diagram shows the contents of the main memory:

Main memory	
800	0110 0100
801	0111 1100
802	1001 0111
803	0111 0011
804	1001 0000
805	0011 1111
806	0000 1110
807	1110 1000
808	1000 1110
809	1100 0010
:	
:	
2000	1011 0101

(c) The program used the ASCII coding system for character codes. An alternative coding system is Unicode.

(i) Give **two** disadvantages of using ASCII code.

1

2

[2]

(ii) Describe how Unicode is designed to overcome the disadvantages of ASCII.

[2]

113. M/J 16/P11/Q2/a,b,c,d

- 2 (a)** Convert the following denary integer into 8-bit binary.

55

--	--	--	--	--	--	--	--

[1]

- (b)** Convert the following Binary Coded Decimal (BCD) number into denary.

10000011

[1]

- (c)** Convert the following denary integer into 8-bit two's complement.

-102

--	--	--	--	--	--	--	--

[2]

- (d)** Convert the following hexadecimal number into denary.

4E

[1]

114. M/J 16/P11/Q4/a,b

- 4** A group of students broadcast a school radio station on a website. They record their sound clips (programmes) in advance and email them to the producer.

(a) Describe how sampling is used to record the sound clips.

[3]

(b) The students use software to compress the sound clips before emailing them.

(i) Circle your chosen method of compression and justify your choice.

Lossy / Lossless

Justification:

[3]

Students also email images to the radio station for use on its website.

These are compressed before sending using run-length encoding (RLE).

(ii) Explain what is meant by run-length encoding.

[3]

(iii) The following diagrams show:

- the denary colour code that represents each colour
- the first three rows of a bitmap image

Colour symbol	Colour code (denary)
B	153
W	255

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	B	B	B	B	B	B	B	B	B	B	W	W	W	B	B	B
1	B	B	B	B	B	B	B	B	B	W	W	W	W	W	W	B
2	B	B	B	B	B	B	B	W	W	W	W	W	W	W	W	W
...																
95																

Show how RLE will compress the first three rows of this image.

Row 1:

Row 2:

Row 3:

[2]

115. M/J 16/P12/Q2/a,b,c,d

- 2 (a)** Convert the following denary integer into 8-bit binary.

55

--	--	--	--	--	--	--	--

[1]

- (b)** Convert the following Binary Coded Decimal (BCD) number into denary.

10000011

[1]

- (c)** Convert the following denary integer into 8-bit two's complement.

-102

--	--	--	--	--	--	--	--

[2]

- (d)** Convert the following hexadecimal number into denary.

4E

[1]

116. M/J 16/P12/Q4/a,b

- 4** A group of students broadcast a school radio station on a website. They record their sound clips (programmes) in advance and email them to the producer.

(a) Describe how sampling is used to record the sound clips.

[3]

(b) The students use software to compress the sound clips before emailing them.

(i) Circle your chosen method of compression and justify your choice.

Lossy / Lossless

Justification:

[3]

Students also email images to the radio station for use on its website.

These are compressed before sending using run-length encoding (RLE).

(ii) Explain what is meant by run-length encoding.

[3]

(iii) The following diagrams show:

- the denary colour code that represents each colour
- the first three rows of a bitmap image

Colour symbol	Colour code (denary)
B	153
W	255

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	B	B	B	B	B	B	B	B	B	B	W	W	W	B	B	B
1	B	B	B	B	B	B	B	B	B	W	W	W	W	W	W	B
2	B	B	B	B	B	B	B	W	W	W	W	W	W	W	W	W
...																
95																

Show how RLE will compress the first three rows of this image.

Row 1:

Row 2:

Row 3:

[2]

117. M/J 16/P13/Q2/a,b,c,d

- 2 (a)** Convert the following 8-bit binary integer into denary.

01001101

[1]

- (b)** Convert the following denary number into Binary Coded Decimal (BCD).

82

[1]

- (c)** Convert the following two's complement integer number into denary.

11001011

[2]

- (d)** Convert the following denary number into hexadecimal. Show your working.

198

[2]

118. O/N 15/P11/Q8/a,b

- 8 (a)** Six computer graphics terms and seven descriptions are shown below.

Draw a line to link each term to its correct description.

Term	Description
Bitmap graphic	Measured in dots per inch (dpi); this value determines the amount of detail an image has
Image file header	Picture element
Image resolution	Image made up of rows and columns of picture elements
Pixel	Image made up of drawing objects. The properties of each object determine its shape and appearance.
Screen resolution	Specifies the image size, number of colours, and other data needed to display the image data
Vector graphic	Number of samples taken per second to represent some event in a digital format
	Value quoted for a monitor specification, such as 1024×768 . The larger the numbers, the more picture elements will be displayed.

[6]

- (b) (i) A black and white image is 512 pixels by 256 pixels.

Calculate the file size of this image in kilobytes (KB) (1 KB = 1024 bytes).
Show your working.

[2]

- (ii) Give a reason why it is important to estimate the file size of an image.

[1]

119. O/N 15/P13/Q8/a,b

8 (a) Six computer graphics terms and seven descriptions are shown below.

Draw a line to link each term to its correct description.

Term	Description
Bitmap graphic	Measured in dots per inch (dpi); this value determines the amount of detail an image has
Image file header	Picture element
Image resolution	Image made up of rows and columns of picture elements
Pixel	Image made up of drawing objects. The properties of each object determine its shape and appearance.
Screen resolution	Specifies the image size, number of colours, and other data needed to display the image data
Vector graphic	Number of samples taken per second to represent some event in a digital format
	Value quoted for a monitor specification, such as 1024×768 . The larger the numbers, the more picture elements will be displayed.

[6]

(b) (i) A black and white image is 512 pixels by 256 pixels.

Calculate the file size of this image in kilobytes (KB) (1 KB = 1024 bytes).
Show your working.

[2]

- (ii) Give a reason why it is important to estimate the file size of an image.

[1]

120. M/J 15/P11/Q1

- 1 (i) Convert the following binary number into hexadecimal.

1 0 1 1 1 0 0 0

[1]

- (ii) Convert the following denary number into BCD format.

9 7

[1]

- (iii) Using two's complement, show how the following denary numbers could be stored in an 8-bit register:

114

--	--	--	--	--	--	--	--

- 93

--	--	--	--	--	--	--	--

[2]

121. M/J 15/P11/Q2/a,c

2 (a) Sound can be represented in a computer in a digital format.

(i) Give the definition of the term sampling.

[1]

(ii) Give **one** reason why 16-bit sampling is used in an audio compact disc (CD).

[1]

(iii) Explain what is meant by the term sampling resolution.

[2]

(iv) Give **one** benefit and **one** drawback of using a higher sampling resolution.

Benefit

Drawback

[2]

(c) Explain the difference between *lossless* and *lossy* data compression techniques.

[3]

122. M/J 15/P12/Q1

- 1 (i) Convert the following binary number into hexadecimal.

1 0 1 1 1 0 0 0

[1]

- (ii) Convert the following denary number into BCD format.

9 7

[1]

- (iii) Using two's complement, show how the following denary numbers could be stored in an 8-bit register:

114

--	--	--	--	--	--	--	--

- 93

--	--	--	--	--	--	--	--

[2]

123. M/J 15/P12/Q2/a,c

2 (a) Sound can be represented in a computer in a digital format.

(i) Give the definition of the term sampling.

[1]

(ii) Give **one** reason why 16-bit sampling is used in an audio compact disc (CD).

[1]

(iii) Explain what is meant by the term sampling resolution.

[2]

(iv) Give **one** benefit and **one** drawback of using a higher sampling resolution.

Benefit

Drawback

[2]

(c) Explain the difference between *lossless* and *lossy* data compression techniques.

[3]

124. M/J 15/P13/Q1/a,b

- 1 (a) (i)** Using two's complement, show how the following denary numbers could be stored in an 8-bit register:

124

--	--	--	--	--	--	--	--

-77

--	--	--	--	--	--	--	--

[2]

- (ii)** Convert the two numbers in **part (a) (i)** into hexadecimal.

124

-77

[2]

- (b)** Binary Coded Decimal (BCD) is another way of representing numbers.

- (i)** Write the number 359 in BCD form.

[1]

- (ii)** Describe a use of BCD number representation.

[2]

Answers Section

1. M/J 26/P11/Q1/a,b,c,d

1(a)(i)	1 mark for each correct value <table border="1" style="width: 100%; border-collapse: collapse; text-align: center;"> <thead> <tr> <th>Character</th><th>Denary</th><th>Binary</th><th>Hexadecimal</th></tr> </thead> <tbody> <tr> <td>☼</td><td>489</td><td>0001 1110 1001</td><td>1E9</td></tr> <tr> <td>@</td><td>64</td><td>0100 0000</td><td>40</td></tr> <tr> <td>Ω</td><td>911</td><td>0011 1000 1111</td><td>38F</td></tr> <tr> <td>Ψ</td><td>5590</td><td>0001 0101 1101 0110</td><td>15D6</td></tr> </tbody> </table>	Character	Denary	Binary	Hexadecimal	☼	489	0001 1110 1001	1E9	@	64	0100 0000	40	Ω	911	0011 1000 1111	38F	Ψ	5590	0001 0101 1101 0110	15D6	4	Correct answers only
Character	Denary	Binary	Hexadecimal																				
☼	489	0001 1110 1001	1E9																				
@	64	0100 0000	40																				
Ω	911	0011 1000 1111	38F																				
Ψ	5590	0001 0101 1101 0110	15D6																				
1(a)(ii)	1 mark for the identification - ASCII 1 mark for each difference to max 2 - There are fewer bits per character // ASCII has 7/8 bits per character, UNICODE has many more - Fewer different characters can be represented // ASCII only represents the Latin alphabet - ASCII Cannot be expanded indefinitely	3	Accept extended ASCII, or any other valid character set. English Language BOD																				
1(b)	107	1	Correct answer only																				
1(c)	1 mark for correct working - showing/working binary subtraction 1 mark for correct answer 0100 1101 Method 1: Change +30 to -30 and add 30 in binary: 0001 1110 Flip bits: 1110 0001 +0000 0001 Add the 1: 1110 0010 Add -30: 0110 1011 1110 0010 (1) 0100 1101 Method 2: Direct Subtraction 0110 1011 -0001 1110 0100 1101 11 1 Carries	2	No marks for converting to denary and back Method 1: Direct subtraction Needs to show the carries for the working mark Method 2: Change +30 to -30 and add Needs to show the conversion and the addition																				
1(d)	1 mark each - Binary is in powers of 2 whilst decimal is in powers of 10 - Any example of values e.g. Kibibyte is 1024 bytes whilst Kilobyte is 1000 bytes	2	Do not allow: Decimal uses the digits 0,1,2,3,4,5,6,7,8,9 binary only uses the digits 0 and 1 The question asks about binary and denary prefixes MP2: Binary is mebibyte, denary is megabyte is ok, binary is mebibyte, denary is kilobyte is not																				

2. M/J 26/P11/Q5/a,b,c

5(a)	1 mark for each point to max 3 - The image is made of pixels - Each pixel has one colour - Each colour has a unique binary code - The code for the colour of each pixel is stored in sequence / order	3	
------	---	---	--

5(b)	1 mark for each correct term - drawing object - property - drawing list A drawing object is an individual element of an image, for example: a line, a square. A property is an attribute of one part of the image, for example: a line colour, the width of a line. A drawing list is all of the elements that are included in the vector graphic.	3	
5(c)	No mark for the choice. 1 mark for each point for a matching justification to max 3 Bitmap image: - The logo might include something like a photograph - Bitmaps can be more detailed/precise/high quality - which may be more attractive/appealing on documents - Bitmaps can be edited more precisely - in case changes need to be made to the logo in future Vector graphic: - A logo should be made of simple shapes which are calculated mathematically - A vector graphic logo can be resized without pixelation - so, it could be used on different documents without losing quality - The file size is smaller than a bitmap image - so, the logo will not use excessive file storage if used in many documents	3	Justification must match the candidate's choice. No mix and match. Accept any sensible justification of their choice Accept other examples such as tee shirts for documents in MP3 and MP8

3. M/J 26/P12/Q1/a,b,c,d

1(a)	1 mark for a correct tick The smallest element of an image that can be one colour	1	Correct answer only
1(b)	1 mark for each correct answer - Image resolution is the number of pixels wide by pixels high that form the image and screen resolution is the number of pixels wide by pixels high that can be displayed on a screen. - If screen resolution is not as high as image resolution the quality of the image is worsened. // Image resolution can be changed, but screen resolution has a physical limit	2	Accept horizontal and/or vertical for wide and/or high Just number of pixels for either or both is NE Do not accept bits or characters for pixels
1(c)	1 mark for a correct answer For example: - File size - File type - Dimensions of the image // image size - Compression type - Data offset – bytes from the beginning of the file to the actual pixel data	1	Mark the first answer given Do not accept resolution or colour depth, they are given in the question Do not accept a description of a file header or metadata Accept height and/or width for dimensions BOD
1(d)(i)	1 mark for: Two // 2	1	Correct answer only
1(d)(ii)	1 mark for each correct point to max 2 - Run Length Encoding (RLE) can be used - A series of consecutive pixels of the same colour // a sequence of the same colour pixel is stored as a single copy of the colour and the number of consecutive occurrences 1 mark for a suitable reference to the given image - RLE: e.g. R2 // W4 // 2R4W // R2W4	3	Take care that the answers are compressing pixels not anything else Do not accept bits or characters or elements etc. for pixels If no reference to the diagram included in the answer, please check the diagram for annotations

4. M/J 26/P12/Q4/a,b

4(a)	1 mark for: The sound wave is measured 44100 times each second // 44100 samples are taken every second	1	Accept appropriate alternative words for measured including 'sampled' Accept 44.1 thousand instead of 44100
4(b)	1 mark each to max 3 - The resolution of 2 bytes (16 bits) will have more bits per sample ... which will result in a larger file size ... but the digital sound will be more accurate to original ... because a greater range/more exact amplitudes can be recorded ... with smaller quantisation errors	3	Must be clear which resolution is being described Allow reverse arguments e.g. The resolution of 8 bits will have fewer bits per sample ... which will result in a smaller file size etc.

5. M/J 26/P12/Q5/a,b

5(a)	1 mark for each correctly completed value <table><thead><tr><th>denary</th><th>unsigned binary</th><th>hexadecimal</th><th>Binary Coded Decimal (BCD)</th></tr></thead><tbody><tr><td>160</td><td>10100000</td><td>A0</td><td>0001 0110 0000</td></tr><tr><td>289</td><td>100100001</td><td>121</td><td>001010001001</td></tr><tr><td>2486</td><td>1001 1011 0110</td><td>9B6</td><td>0010010010000110</td></tr><tr><td>35123</td><td>1000100100110011</td><td>8933</td><td>00110101000100100011</td></tr></tbody></table>	denary	unsigned binary	hexadecimal	Binary Coded Decimal (BCD)	160	10100000	A0	0001 0110 0000	289	100100001	121	001010001001	2486	1001 1011 0110	9B6	0010010010000110	35123	1000100100110011	8933	00110101000100100011	4	
denary	unsigned binary	hexadecimal	Binary Coded Decimal (BCD)																				
160	10100000	A0	0001 0110 0000																				
289	100100001	121	001010001001																				
2486	1001 1011 0110	9B6	0010010010000110																				
35123	1000100100110011	8933	00110101000100100011																				
5(b)	1 mark for each correct answer <table><thead><tr><th>statement</th><th>answer</th></tr></thead><tbody><tr><td>The smallest 12-bit two's complement binary number</td><td>1000 0000 0000</td></tr><tr><td>The denary value represented by the 8-bit unsigned binary number 01100010 after a logical right shift of 2 places has been applied.</td><td>24</td></tr><tr><td>The binary value after a cyclic left shift of 4 places has been applied to the binary number 10010101</td><td>0101 1001</td></tr></tbody></table>	statement	answer	The smallest 12-bit two's complement binary number	1000 0000 0000	The denary value represented by the 8-bit unsigned binary number 01100010 after a logical right shift of 2 places has been applied.	24	The binary value after a cyclic left shift of 4 places has been applied to the binary number 10010101	0101 1001	3	Note: The first answer must be 12 bits and a binary value Note: The second answer must be in denary.												
statement	answer																						
The smallest 12-bit two's complement binary number	1000 0000 0000																						
The denary value represented by the 8-bit unsigned binary number 01100010 after a logical right shift of 2 places has been applied.	24																						
The binary value after a cyclic left shift of 4 places has been applied to the binary number 10010101	0101 1001																						

6. M/J 26/P13/Q3/a,b,c(i)

3(a)	1 mark for: Each sample is stored as a 16-bit binary number // there are 2^{16} / 65536 different amplitudes that can be recorded	1	Accept 2 bytes Or any equivalent.
3(b)(i)	1 mark each to max 3 - Image is a series of pixels ... where each pixel has one colour - Each colour has a unique binary number - The binary number/code of the colour of each pixel is stored in sequence	3	
3(b)(ii)	1 mark for working 1 mark for answer $2048 * 1536 * 2 * 8 / 1024 / 1024$ 48 MiB	2	

3(c)(i)	1 mark each max 2 for image and max 2 for sound Image e.g. - Run-length encoding / RLE - Store each colour and the number of times it appears consecutively Sound e.g. - Reduce amplitude values (dynamic range) to only those sounds used in the actual recording reducing sampling resolution without impacting the sound - Record changes in amplitudes instead of each individual amplitudeeach change requires less bits than a new amplitude	4	Accept RLE for sound but do not allow for both parts
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7. M/J 26/P13/Q6/a,b,c,d

6(a)	1 mark for each set of lines from each denary number <table><thead><tr><th>Denary number</th><th>Equivalent unsigned value</th></tr></thead><tbody><tr><td>3120</td><td>Hexadecimal: 121</td></tr><tr><td>289</td><td>BCD: 00010101100001010001</td></tr><tr><td>22516</td><td>Binary: 110000110000</td></tr><tr><td>15851</td><td>Hexadecimal: 57F4</td></tr><tr><td></td><td>Binary: 11110111101011</td></tr><tr><td></td><td>BCD: 0011000100100000</td></tr></tbody></table>	Denary number	Equivalent unsigned value	3120	Hexadecimal: 121	289	BCD: 00010101100001010001	22516	Binary: 110000110000	15851	Hexadecimal: 57F4		Binary: 11110111101011		BCD: 0011000100100000	4	
Denary number	Equivalent unsigned value																
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	Binary: 11110111101011																
	BCD: 0011000100100000																
6(b)	1 mark for working 1 mark for answer. 1101 0100 1111 1111 1100 1111 1 1101 0001 1110 1111 1 1 111	2	No marks for answer if no evidence of using binary addition.														
6(c)	1 mark for correct answer 1713	1															
6(d)	1 mark for correct answer 1000000000000000	1	Last answer														

8. O/N 25/P11/Q1/a,b

1(a)(i)	B3A	1
1(a)(ii)	0001 0000 1000	1

1(a)(iii)	1 mark for the working 1 mark for the correct denary value Working: Method 1: <pre> 1111 1011 1100 Flip the bits 0000 0100 0011 Add 1 0000 0000 0001 + 0000 0100 0100 </pre> Method 2: -2048 + 1024 + 512 + 256 + 128 + 32 + 16 + 8 + 4 // -128 + 32 + 16 + 8 + 4 Denary value: -68	2
1(b)(i)	(1) 00101011	1
1(b)(ii)	1 mark per bullet point, max 2 marks - An overflow error occurs - The answer cannot be represented in the number of bits available // The answer is larger than the maximum positive number that can be stored in the register // The answer is smaller than the most negative number that can be stored in the register	2

9. O/N 25/P11/Q7/d

7(d)(i)	1 mark for the working 1 mark for the correct answer Working: $(512 * 2048 * 8) / (8 * 1024) // (512 * 2048) / 1024$ $// (2^9 * 2^{11} * 2^3) / (2^3 * 2^{10}) // (2^9 * 2^{11}) / 2^{10}$ Answer: 1024 kibibytes // 2^{10} kibibytes	2
7(d)(ii)	1 mark per bullet point, max 2 marks - The file will contain metadata - The file will have a header	2
7(d)(iii)	1 mark per bullet point, max 2 marks - Sequences of consecutive identical colours / pixels ... are stored as the colour value and the number of times it occurs consecutively	2

10. O/N 25/P12/Q6/a,b,d

6(a)	1 mark for correct answer Type of compression: lossy Justification: There will be fewer samples per second, so data will be permanently lost // There will be fewer samples per second, so the original sound cannot be re-created	1								
6(b)	1 mark for correct words in the correct order <table><tr><td>Binary Value</td><td>0011 1000</td><td>0011 1100</td><td>0011 1110</td></tr><tr><td>Word</td><td>Science</td><td>is</td><td>Amazing!</td></tr></table>	Binary Value	0011 1000	0011 1100	0011 1110	Word	Science	is	Amazing!	1
Binary Value	0011 1000	0011 1100	0011 1110							
Word	Science	is	Amazing!							
6(d)	1 mark for the working 1 mark for the correct answer Working: (1000 * 2000 * 16) / (8 * 1000 * 1000) // (1000 * 2000 * 2 / (1000 * 1000) Answer: 4 megabytes	2								

11. O/N 25/P12/Q7/a,b,c

7(a)	1 mark for the correct answer 1100 0000	1
7(b)	1 mark for the correct answer 8 // 16 // 32 // 64	1
7(c)(i)	1 mark for the correct answer 0001 0000 0100	1
7(c)(ii)	1 mark for the correct answer 68	1

12. O/N 25/P13/Q1/a,b,c

1(a)	1 mark for each correctly connected change box, max 3 marks	3
1(b)	1 mark per bullet point, max 2 marks • Each character has a unique code • Each character in the text is replaced sequentially / in order by its code	2
1(c)	1 mark per bullet point, max 2 marks • ASCII uses 7 / 8 bits Unicode can use many more / up to 32 bits • Unicode can represent a wider range of characters including different languages	2

13. O/N 25/P13/Q2/a,b,d

2(a)	3 marks for 6 correct ticks 2 mark for 4 or 5 correct ticks 1 mark for 2 or 3 correct ticks	3																																																															
	<table><tr><th rowspan="2">Example of data</th><th colspan="7">Number of bits</th></tr><tr><th>4</th><th>8</th><th>16</th><th>24</th><th>32</th><th>64</th><th>128</th></tr><tr><td>the hexadecimal value F139</td><td></td><td></td><td>✓</td><td></td><td></td><td></td><td></td></tr><tr><td>16 000 000 unique amplitude values</td><td></td><td></td><td></td><td>✓</td><td></td><td></td><td></td></tr><tr><td>an IPv4 address</td><td></td><td></td><td></td><td></td><td>✓</td><td></td><td></td></tr><tr><td>256 unique colours</td><td></td><td>✓</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>an IPv6 address</td><td></td><td></td><td></td><td></td><td></td><td></td><td>✓</td></tr><tr><td>the denary value 65 000</td><td></td><td></td><td>✓</td><td></td><td></td><td></td><td></td></tr></table>	Example of data	Number of bits							4	8	16	24	32	64	128	the hexadecimal value F139			✓					16 000 000 unique amplitude values				✓				an IPv4 address					✓			256 unique colours		✓						an IPv6 address							✓	the denary value 65 000			✓					
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2(b)	1 mark 1111 1001 0100	1																																																															
2(d)	1 mark E33B	1																																																															

14. O/N 25/P13/Q7/b

7(b)	1 mark for the working 1 mark for the correct answer Working: $(4000 \times 3000 \times 30 \times 16) / (8 \times 1000 \times 1000 \times 1000)$ $// (4000 \times 3000 \times 30 \times 2) / (1000 \times 1000 \times 1000)$ Answer: 0.72 gigabytes	2
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15. M/J 25/P11/Q2/b

2(b)(i)	1 mark each to max 4 e.g. • Video is data-intensive • The file size needs reducing in order to • reduce the amount of bandwidth used • and reduce buffering • This means people are not behind in the conversation • and people with lower bandwidth can still take part	4
2(b)(ii)	No mark for choice but lossy is most appropriate 1 mark each to max 3 e.g. • Reduces file size more than lossless • so significantly less bandwidth / data is needed • so buffering is reduced even more than with lossless • Data can be removed which cannot be seen • reducing quality without impacting experience • for example, because resolution of video can be reduced // sample rate of audio can be reduced	3

16. M/J 25/P11/Q3/a,b

3(a)	1 mark for each correct answer <table><tr><th>Statement</th><th>Answer</th></tr><tr><td>The term for the smallest element that makes up an image.</td><td>pixel</td></tr><tr><td>The largest number of different colours that can be represented with a bit depth of 8 bits.</td><td>256 // 2⁸</td></tr><tr><td>The term for the dots per inch (dpi) when an image is displayed.</td><td>Screen resolution</td></tr></table>	Statement	Answer	The term for the smallest element that makes up an image.	pixel	The largest number of different colours that can be represented with a bit depth of 8 bits.	256 // 2⁸	The term for the dots per inch (dpi) when an image is displayed.	Screen resolution	3
Statement	Answer									
The term for the smallest element that makes up an image.	pixel									
The largest number of different colours that can be represented with a bit depth of 8 bits.	256 // 2⁸									
The term for the dots per inch (dpi) when an image is displayed.	Screen resolution									
3(b)(i)	1 mark each to max 2 • A wider range of characters can be represented • so characters from more languages can be represented • and symbols such as emojis can be used	2								
3(b)(ii)	1 mark for: 10,094	1								
3(b)(iii)	1 mark for: 8,512	1								

17. M/J 25/P12/Q2/a,b,c

2(a)	1 mark for: 0010 0010 1110 1 mark for: 22E	2
2(b)(i)	1 mark for: -30	1
2(b)(ii)	1 mark for smallest, 1 mark for largest Smallest: 1000 0000 Largest: 0111 1111	2
2(c)	1 mark for application and 1 mark for corresponding justification e.g. - an application that performs financial / banking calculations - because financial transactions use only two decimal places and must be accurate, no accumulating / rounding errors and it is difficult to represent decimal values exactly in normal binary. Or - electronic displays - because visual displays only need to show individual digits - because conversion between denary and BCD is straightforward Or - The storage of the date and time in the BIOS of a PC - because conversion between denary and BCD is more straightforward Or - Barcode systems - because conversion between denary and BCD can be accurately completed	2

18. M/J 25/P13/Q1/a,b,c

1(a)(i)	1 mark for correct working: e.g. $2000000 * 16 / (8 * 1000 * 1000)$ 1 mark for answer: 4MB	2
1(a)(ii)	1 mark each Image: - There will be fewer shades of colour available - so the image does not match the original as detail is lost Image file: - Fewer bits are used to store each pixel - so less data is stored, therefore the file size is reduced	4

1(b)	1 mark for each correct term: <table><tr><th>Description</th><th>Sound term</th></tr><tr><td>the number of times the amplitude is measured per time interval</td><td>Sampling rate</td></tr><tr><td>the number of bits used to store each amplitude measurement</td><td>Sampling resolution</td></tr><tr><td>the type of sound wave before it is recorded by a computer</td><td>Analogue</td></tr></table>	Description	Sound term	the number of times the amplitude is measured per time interval	Sampling rate	the number of bits used to store each amplitude measurement	Sampling resolution	the type of sound wave before it is recorded by a computer	Analogue	3								
Description	Sound term																	
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the number of bits used to store each amplitude measurement	Sampling resolution																	
the type of sound wave before it is recorded by a computer	Analogue																	
1(c)(i)	1 mark each to max 2 8 / 16 / 32 / bits per character - Represents $2^8 / 2^{16}$ / etc. characters - Represents every language and other characters such as emojis	2																
1(c)(ii)	1 mark for each correctly completed space <table><tr><th>Character</th><th>Denary</th><th>8-bit Binary</th><th>Hexadecimal</th></tr><tr><td>!</td><td>33</td><td>0010 0001</td><td>21</td></tr><tr><td>L</td><td>76</td><td>0100 1100</td><td>4C</td></tr><tr><td>ü</td><td>252</td><td>1111 1100</td><td>FC</td></tr></table>	Character	Denary	8-bit Binary	Hexadecimal	!	33	0010 0001	21	L	76	0100 1100	4C	ü	252	1111 1100	FC	3
Character	Denary	8-bit Binary	Hexadecimal															
!	33	0010 0001	21															
L	76	0100 1100	4C															
ü	252	1111 1100	FC															

19. M/J 25/P13/Q7/b,c

7(b)	1 mark for $ \begin{array}{r} 1\ 1\ 1\ 1\ 0\ 1\ 0\ 1 \\ +_1\ 1\ 1\ 0\ 1\ 1\ 0\ 0\ 0\ 1\ 1 \\ \hline (1)\ 1\ 0\ 1\ 0\ 0\ 1\ 1\ 0 \end{array} $ Answer: (1) 1010 0110	1
7(c)	1 mark for showing binary subtraction (any method) Direct subtraction: $ \begin{array}{r} 0\ 1\ 1\ 1\ 0\ 1\ 1\ 0\ 0 \\ -0\ 0\ 0\ 1\ 1\ 1\ 1\ 1\ 0 \\ \hline 0\ 1\ 0\ 0\ 0\ 1\ 1\ 0 \end{array} $ Adding the two's complement: $ \begin{array}{r} 0\ 1\ 1\ 0\ 0\ 1\ 0\ 0 \\ +_1\ 1\ 1\ 1\ 1\ 0\ 0\ 0\ 1\ 0 \\ \hline (1)\ 0\ 1\ 0\ 0\ 0\ 1\ 1\ 0 \end{array} $ 1 mark for answer 0100 0110	2

20. O/N 24/P11/Q1/a,b,c,d

1(a)	1 mark for: - A tebibyte = 1024 gibibytes / 1 048 576 kibibytes / 2^{40} bytes whereas a gigabyte = 1000 megabytes / 1 000 000 kilobytes / 10^9 bytes - Tebi is binary prefix giga is denary prefix	1
1(b)(i)	1 mark for: C67	1
1(b)(ii)	1 mark for: -1641	1
1(b)(iii)	1 mark for: 573	1
1(c)	1 mark for working 1 mark for answer. Working Using the two's complement and adding: 23 in binary = 00010111 so minus 23 is 11101001 0100 1010 + <u>1110 1001</u> (1)0011 0011 Direct subtraction: 0100 1010 - <u>0001 0111</u> 0011 0011 Answer: 0011 0011	2
1(d)	1 mark from: - The answer cannot be represented in the number of bits available - The answer is larger than the maximum positive number that can be stored in the register - The answer is smaller than the most negative number that can be stored in the register	1

21. O/N 24/P11/Q4/b

4(b)(i)	1 mark for each bullet point (max 3) e.g. • Less of the teacher's storage space is used • ... so more student work can be stored • Transmission time is reduced • ... so the teacher does not have to wait as long for it to arrive • Bandwidth usage is reduced • ... so other transmissions are not adversely affected • Less data allowance is used on the teacher's email system	3
4(b)(ii)	1 mark for each bullet point (max 3) • Run-Length Encoding // RLE • Repeated sequences of the same characters are replaced by • ... a single copy of the character • ... and a count of the number of characters	3

22. O/N 24/P12/Q7/a,b

7(a)(i)	1 mark for each correct definition Colour depth: • the number of bits used to represent a colour // the number of colours that can be represented in an image File header: • stores data about the image file / metadata	2
7(a)(ii)	1 mark for each correct explanation Image quality: • Decreasing resolution means details within the image are lost because there are fewer pixels // Increasing resolution means the image is more detailed because there are more pixels File size: • Decreasing the resolution will decrease the file size because there are fewer pixels therefore less data // Increasing the resolution will increase the file size because there are more pixels therefore more data	2
7(a)(iii)	1 mark for correct method For example: • Run-Length Encoding	1

7(b)	1 mark for each correct definition Property: - an attribute of a drawing object // data about a shape // defines one aspect of the appearance of a drawing object Drawing list: - all the drawing objects/shapes in an image // stores the commands/descriptions / mathematical equations required to draw each object	2
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23. O/N 24/P13/Q6/a,b,c

6(a)	1 mark for identification 1 mark for matching description e.g. - ASCII 7/8 bits per character // represents 128/256 characters // represents all characters from Latin alphabet - UNICODE 8/16/32 bits per character // represents 256/65536+ characters // represents all characters in all languages	2
6(b)(i)	1 mark for: 256 // 2^8	1
6(b)(ii)	1 mark for: Increased file size	1
6(b)(iii)	1 mark for each bullet point (max 2) e.g. - The change may not be noticeable // Data removed is usually not noticed by the human eye ... for example, changes in shade/detail - It produces a larger decrease in file size compared to lossless // Lossy decreases file size considerably	2
6(c)(i)	1 mark for each bullet point (max 2) - Value/magnitude/size of the analogue sound wave is measured a set number of times each second/time / at set intervals - Each sample/reading/measurement is given the binary number and stored in sequence	2

6(c)(ii)	1 mark for each correct point and 1 mark for matching expansion e.g. • Decrease sample rate • ... fewer samples/readings/measurements stored per second // fewer bits per second stored • Decrease sample resolution • ... fewer bits per sample/reading/measurement // each sample has fewer bits • Sound outside of set/human hearing range is removed • ... fewer measurements are stored / decreases the number of possible binary values so fewer bits are stored	2
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24. O/N 24/P13/Q8/a,b

8(a)	1 mark for: Denary value: 8107	1
8(b)	1 mark for each bullet point for the method (max 2) e.g. • Flip each bit then add 1 ... • ... method of converting the new binary number into denary • Most significant 1 bit is treated as the corresponding negative denary value ... • ... add the other positive corresponding denary values 1 mark for correct conversion Denary value: -97	3

25. M/J 24/P11/Q2/d

2(d)	1 mark each to max 3 : • Data transmission to user's smartphone will take longer • ... because there is more data to transmit • The secondary storage device will fill faster • ... fewer videos will be able to be stored long-term // videos are overwritten more often	3
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26. M/J 24/P11/Q7

7	1 mark each: - Working – carried values clearly indicated - Correct answer 0001 1000 - Overflow clearly indicated as overflow Example: <pre> 1 0 0 1 1 1 1 0 0 1 1 0 0 0 0 1 + 0 0 0 1 1 0 0 1 ----- (1) 0 0 0 1 1 0 0 0 1 1 1 1 1 1 1(carries) </pre>	3
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27. M/J 24/P12/Q2/d

2(d)(i)	1 mark each: - Image is made of pixels and each pixel has one colour - Each colour has a unique binary code Code for the colour of each pixel is stored in sequence	3
2(d)(ii)	1 mark each to max 2: - List of objects in the drawing - A list that stores the command/description/equation required to draw each object - Properties of each object e.g. the fill colour, line weight/colour	2
2(d)(iii)	1 mark each to max 2: - Dedicated connection to the headset // not sharing bandwidth - Already fast connection that can transmit the data without slowing - Video may already be a small file size and does not need further reduction - Video is not saved so storage is not an issue in the headset	2

28. M/J 24/P12/Q7/a,b,c

7(a)	1 mark for: 3300 kibibytes	1
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7(b)	1 mark each: - Converting 100 to binary 0110 0100 and 10 to binary 0000 1010 - Subtraction method - converting 10 to – 10 and adding // direct subtraction correct answer 0101 1010 Method 1: Converting to -10 and adding: Binary for +10 is 0000 1010 Binary for –10 is 1111 0110 Binary for 100 is 0110 0100 100 + (-10): <pre> 0 1 1 0 0 1 0 0 +1 1 1 1 0 1 1 0 (1) 0 1 0 1 1 0 1 0 Carries: 1 1 0 0 1 0 0 0 </pre> Method 2: Direct Subtraction Borrows: <pre> 0 0 0 1 1 0 1 0 0 1 1 0 0 1 0 0 - 0 0 0 0 1 0 1 0 0 1 0 1 1 0 1 0 </pre>	3
7(c)	1 mark for working: 1100 0000 1111 // 2048 + 1024 + 8 + 4 + 2 + 1 // (12 * 16²) + 15 // (12 * 16 * 16) + 15 // 3072 + 15 1 mark for correct answer: 3087	2

29. M/J 24/P13/Q1/a,b,c,d

1(a)	1 mark for each correct answer: - binary 3072 - denary/decimal 2000 A kibibyte has a binary prefix. Three kibibytes is the same as 3072 bytes. A megabyte has a decimal/denary prefix. Two terabytes is the same as 2000 gigabytes.	4
1(b)	1 mark for correct answer: F1	1

1(c)	1 mark for a correct answer: The answer is too long to be represented in the same number of bits as the binary numbers being added	1								
1(d)(i)	1 mark for all 3 answers correct: <table><tr><th>Character set</th><th>Number of bits</th></tr><tr><td>ASCII</td><td>7</td></tr><tr><td>extended ASCII</td><td>8</td></tr><tr><td>Unicode</td><td>16/32</td></tr></table>	Character set	Number of bits	ASCII	7	extended ASCII	8	Unicode	16/32	1
Character set	Number of bits									
ASCII	7									
extended ASCII	8									
Unicode	16/32									
1(d)(ii)	1 mark each: - Each character has a unique binary code - The binary code for each character is stored in sequence	2								

30. M/J 24/P13/Q2/a,b,c

2(a)	1 mark for working: $4000 * 3000 * 4$ 1 mark for correct answer: 48MB	2
2(b)(i)	1 mark each to max 3 : - The file takes less storage space on the web server than if lossless compression was used - The file is faster to upload/download to/from the server than if lossless compression was used - The file uses less bandwidth to transmit than if lossless compression was used - The file consumes less data allowance than if lossless compression was used	3
2(b)(ii)	1 mark each: - Identifies consecutive repeating pixels of the same colour - Stores the colour /pattern and the number of times it repeats	2
2(c)	1 mark each to max 2 : - Colour/bit depth - Image resolution	2

31. O/N 23/P11/Q1/a,b,c

1(a)	2 marks for all 3 lines correct 1 mark for 1 line correct <table><thead><tr><th>Term</th><th>Description</th></tr></thead><tbody><tr><td>drawing list</td><td>a component created using a formula</td></tr><tr><td>drawing object</td><td>defines one characteristic of a component</td></tr><tr><td>property</td><td>data required to create all components in the graphic</td></tr></tbody></table>	Term	Description	drawing list	a component created using a formula	drawing object	defines one characteristic of a component	property	data required to create all components in the graphic	2
Term	Description									
drawing list	a component created using a formula									
drawing object	defines one characteristic of a component									
property	data required to create all components in the graphic									
1(b)	1 mark for the definition - The number of bits used to represent each colour 1 mark for each bullet point for the explanation - Increase in bit depth means the image has a greater range of colours // Decrease in bit depth means the image has a smaller range of colours - Increase in bit depth makes the image closer to the original / more realistic // Decrease in bit depth makes the image less like the original / less realistic	3								
1(c)	1 mark for each bullet point (max 2) - Reduced bandwidth usage when transmitting the message - Reduced transmission time from email client to email server - Reduced storage space on the email - Email accounts often have a maximum size for an attachment	2								

32. O/N 23/P12/Q3/a,b,c

3(a)	1 mark for: Either - kibibyte = 1,024 bytes // 2^{10} bytes and megabyte = 1000 kilobytes // 1 000 000 bytes // 10^3 kilobytes // 10^6 bytes Or - kibi is binary prefix and mega is denary prefix	1
3(b)(i)	1111 0011 1100	1
3(b)(ii)	865	1
3(b)(iii)	470	1

3(c)	1 mark for a correct application 1 mark for a corresponding justification - An application that performs financial / banking calculations ... because it is difficult to represent decimal values exactly in normal binary and financial transactions use only two decimal places and must be accurate, no accumulating errors Or - Electronic displays, e.g. calculators, digital clocks ... because visual displays only need to show individual digits ... because conversion between denary and BCD is easier Or - The storage of the date and time in the BIOS of a PC ... because conversion with denary is easier	2
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33. O/N 23/P12/Q6/a,b,c

6(a)	1 mark for each bullet point (max 3) Lossy compression (ticked) - Loss of quality will not be noticed - Needs to be viewed in real time so less bandwidth needed if file size smaller - Smaller file sizes will reduce buffering so the video will play more smoothly - Viewers may watch on different devices, so may not need high quality resolution Lossless compression (ticked) - Original recording may not have been made in high resolution - Could be streaming to high bandwidth devices - The reduction in the file size is sufficient for the receiving device - Viewers do not want any loss of quality	3
6(b)	1 mark for each bullet point (max 3) Increase sampling resolution ... the number of bits used for each sample is increased ... there will be more values available to represent each sample // more amplitudes can be represented ... each binary amplitude/note in the digital recording is closer to the analogue amplitude/note ... quantisation errors are reduced ... the digital soundwave is closer to the original analogue soundwave Decrease sampling resolution ... the number of bits used for each sample is decreased ... there will be fewer values available to represent each sample // fewer amplitudes can be stored ... each binary amplitude/note in the digital recording is further from the analogue amplitude/note ... quantisation errors are increased ... the digital soundwave is less like the original analogue soundwave	3

6(c)	1 mark for answer, 1 mark for working Answer: 2.5 mebibytes Working: $(2048 \times 1024 \times 10) / (8 \times 1024 \times 1024)$	2
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34. O/N 23/P13/Q1/a,b

1(a)	1 mark for: data values that are continuously changing // variable // any value	1								
1(b)	1 mark for 1 correct line 2 marks for 3 correct lines <table><thead><tr><th>Term</th><th>Description</th></tr></thead><tbody><tr><td>sampling</td><td>the number of samples taken per second</td></tr><tr><td>sampling rate</td><td>taking measurements at regular intervals and storing the values</td></tr><tr><td>sampling resolution</td><td>the number of bits used to store each sample</td></tr></tbody></table>	Term	Description	sampling	the number of samples taken per second	sampling rate	taking measurements at regular intervals and storing the values	sampling resolution	the number of bits used to store each sample	2
Term	Description									
sampling	the number of samples taken per second									
sampling rate	taking measurements at regular intervals and storing the values									
sampling resolution	the number of bits used to store each sample									

35. O/N 23/P13/Q2/a,b

2(a)	1 mark for each bullet point: - more pixels can be stored / are available - the image is sharper / less pixelated	2
2(b)	1 mark for answer, 1 mark for working Answer: 4096 kibibytes Working: File size = $(2048 \times 1024 \times 16) / (8 \times 1024)$	2

36. O/N 23/P13/Q5/b(i)

5(b)(i)	1 mark for each bullet point (max 2). - To reduce the time it takes to download the program files from the web server // to upload them to the server in the first place - To reduce the amount of storage space used on the web server // user's device	2
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37. M/J 23/P11/Q1/a,b,c

1(a)	1 mark for each correct definition <table><tr><th>Term</th><th>Definition</th></tr><tr><td>Drawing list</td><td>All the drawing objects in an image // a list that stores the commands required to draw each object</td></tr><tr><td>Pixel</td><td>The smallest part of the image // one square / dot of one colour</td></tr><tr><td>Colour depth</td><td>The number of bits per pixel // determines the number of colours that can be represented in the image</td></tr></table>	Term	Definition	Drawing list	All the drawing objects in an image // a list that stores the commands required to draw each object	Pixel	The smallest part of the image // one square / dot of one colour	Colour depth	The number of bits per pixel // determines the number of colours that can be represented in the image	3
Term	Definition									
Drawing list	All the drawing objects in an image // a list that stores the commands required to draw each object									
Pixel	The smallest part of the image // one square / dot of one colour									
Colour depth	The number of bits per pixel // determines the number of colours that can be represented in the image									
1(b)(i)	1 mark each Examples: • Confirmation that it is a bitmap // file type • Compression type • Location/offset of data within the file • Dimensions e.g.100 × 100 pixels	2								
1(b)(ii)	1 mark for working; 1 mark for answer • Working: (1500 * 3000 * 8) / 1000 / 1000 • Answer: 36 MB	2								
1(c)(i)	1 mark each Examples: • The customers will be able to download the photographs in less time • ...and they will take less of the customer's bandwidth • The photographs will take up less space on the customer's storage medium • ...therefore the customers can store more images • ...and will have more space for other files	3								

1(c)(ii)	1 mark each to max 2 for explanation; 1 mark for an image related example - RLE stores a colour and the number of times it occurs consecutively - An image may not have many sequences of the same colour - It would need to store each colour and then the count/number 1 which adds data Example: - Red-Green-Blue would become Red 1 Green 1 Blue 1	3
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38. M/J 23/P11/Q3/d(i,ii,iii,iv,v)

3(d)(i)	1 mark from - Kibibyte is 1024 bytes and kilobyte is 1000 bytes - Kibibyte is binary prefix and kilobyte is denary prefix	1
3(d)(ii)	1001 0110 0100	1
3(d)(iii)	F2	1
3(d)(iv)	Smallest: 10000000 Largest: 01111111	2
3(d)(v)	1 mark for working 1 mark for answer <pre> 1 0 1 1 0 0 0 0 + 0 0 0 1 1 0 1 1 1 1 0 0 1 0 1 1 1 1 </pre>	2

39. M/J 23/P12/Q4/a,b,c

4(a)	2^{16} // 65536	1
4(b)	1 mark for working; 1 mark for answer - Working: +120 = 0111 1000 - Answer: 1000 0111	2
4(c)	1 mark for working; 1 mark for answer - Working: $A04 = (10 * 16^2) + 4$ // $A04 = (10 * 256) + 4$ // $A04 = 1010 0000 0100$ - Answer: 2564	2

40. M/J 23/P13/Q3/c

3(c)(i)	1 mark each • The amplitude is recorded a set number of times a second • Each (instance of an) amplitude is given a corresponding binary number • The binary number (of each amplitude) is saved in sequence	3
3(c)(ii)	1 mark each; max 2 for rate and max 2 for resolution Sampling rate • There are smaller 'gaps' in the sound wave // sound is recorded more often • Digital waveform is closer to the analogue waveform • The quantisation errors are smaller Sampling resolution • There are more bits per sample // a wider range of amplitudes can be stored • Each binary amplitude/note (in the digital recording) is closer to the analogue amplitude/note • Digital waveform is closer to the analogue waveform • The quantisation errors are smaller	4

41. M/J 23/P13/Q7/a

7(a)	1 mark for each correct line	3
Description		Denary value
The smallest integer that can be represented in 8-bit two's complement.		-127
		127
The largest integer that can be represented in 8-bit two's complement.		-255
		-128
The largest unsigned integer that can be represented in 8 bits.		-256
		256
		128
		255

42. O/N 22/P11/Q1/a,b,c,d

1(a)(i)	39	1
1(a)(ii)	27	1
1(a)(iii)	-25	1

1(b)	1 mark for working, 1 mark for answer (0011 1110) Working using borrowing: <pre> 1 10 10 1 0 10 0 0 10 10 1 0 1 1 0 0 1 1 0 1 1 1 0 1 0 1 - ----- 0 0 1 1 1 1 1 0 </pre> Working using two's complement: <pre> 1 0 1 1 0 0 1 1 Two's complement 1 0 0 0 1 0 1 1 + (1) 0 0 1 1 1 1 1 0 1 1 </pre>	2
1(c)	1 mark for similarity, 2 marks for differences Similarity (max 1): • both can use 8 bits • both represent each character using a unique code • Unicode will contain all the characters that ASCII contains // ASCII is a subset of Unicode Differences (max 2): • Unicode can go up to 32 bits per character whereas ASCII is 7 or 8 bits Unicode can represent a wider range of characters than ASCII • different languages are represented using Unicode, ASCII is only for one language	3
1(d)(i)	the number of samples taken per unit time / per second	1
1(d)(ii)	1 mark for each bullet point (max 2): • increases the number of bits per sample // larger range of values • which means that the file size increases • makes the sound file more accurate //digital waveform closer to original (analogue) waveform • smaller quantisation errors	2

43. O/N 22/P11/Q7/c

7(c)	1 mark for each bullet point (max 3): • lossless compression • Run Length Encoding • repeated sequences of <u>characters</u> are replaced by • ... a single copy of the character • ... and a counter of the number of characters	3
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44. O/N 22/P12/Q2/a,b

2(a)(i)	−106	1
2(a)(ii)	96	1
2(a)(iii)	1 mark for each bullet point: • 149 decimal • 0001 0100 1001	2
2(b)	1101 0010	1

45. O/N 22/P12/Q6/b

6(b)(i)	1 mark for each bullet point (max 2): • improves the accuracy of the sound file • ... because (digital) waveform more closely resembles the analogue waveform • quantization errors are reduced • increases the amount of detail stored	2
6(b)(ii)	1 mark for each bullet point: • decreases the file size of the sound file • ... because fewer bits are used to store each sample	2

46. O/N 22/P12/Q8/a,b,c

8(a)	1 mark one or two correct row(s). 2 marks for all three correct rows.				2
	Action	Increases the file size	Decreases the file size	No change to the file size	
	Change the colour depth of the image file to 16 bits per pixel.		✓		
	Change the screen resolution to 1366 × 768 pixels.			✓	
	Change the colour of the rectangle from black to red.			✓	

8(b)	1 mark for each bullet point (max 2). For example: • can be enlarged without pixelation / loss of quality • individual components of the image can be edited • generally a smaller file size	2								
8(c)(i)	1 mark for each correct underlined part: <table><tr><th>Uncompressed sound</th><th>RLE compressed sound</th></tr><tr><td>EA F1 F1 F2 F2 F2 EA</td><td>1EA 2F1 3F2 1EA</td></tr><tr><td><u>AB AB FF FF 1D 67</u></td><td>2AB 2FF 11D 167</td></tr><tr><td>32 32 80 81 81</td><td><u>232 180 281</u></td></tr></table>	Uncompressed sound	RLE compressed sound	EA F1 F1 F2 F2 F2 EA	1EA 2F1 3F2 1EA	<u>AB AB FF FF 1D 67</u>	2AB 2FF 11D 167	32 32 80 81 81	<u>232 180 281</u>	2
Uncompressed sound	RLE compressed sound									
EA F1 F1 F2 F2 F2 EA	1EA 2F1 3F2 1EA									
<u>AB AB FF FF 1D 67</u>	2AB 2FF 11D 167									
32 32 80 81 81	<u>232 180 281</u>									
8(c)(ii)	1 mark for each bullet point: • all the data is required // no data can be lost • ... otherwise text file will be corrupted / not make sense	2								

47. O/N 22/P13/Q1/a,b

1(a)	1 mark for one or two correct ticks, 2 marks for three correct ticks.	2																
	<table><tr><th>Action</th><th>Accuracy increases</th><th>Accuracy decreases</th><th>Accuracy does not change</th></tr><tr><td>Change the sampling rate from 40 kHz to 60 kHz.</td><td>✓</td><td></td><td></td></tr><tr><td>Change the duration of the recording from 20 minutes to 40 minutes.</td><td></td><td></td><td>✓</td></tr><tr><td>Change the sampling resolution from 24 bits to 16 bits.</td><td></td><td>✓</td><td></td></tr></table>	Action	Accuracy increases	Accuracy decreases	Accuracy does not change	Change the sampling rate from 40 kHz to 60 kHz.	✓			Change the duration of the recording from 20 minutes to 40 minutes.			✓	Change the sampling resolution from 24 bits to 16 bits.		✓		
Action	Accuracy increases	Accuracy decreases	Accuracy does not change															
Change the sampling rate from 40 kHz to 60 kHz.	✓																	
Change the duration of the recording from 20 minutes to 40 minutes.			✓															
Change the sampling resolution from 24 bits to 16 bits.		✓																
1(b)	1 mark for answer; 1 mark for working. Working: Size = 50KHz * (20 × 60) * 16 bits = 50 000 * 1200 * 16 bits // 50 000 *1200 * 2 bytes = 960 000 000 bits = 120 000 000 bytes = 120 000 kilobytes = 120 megabytes Answer = 120 megabytes	2																

48. O/N 22/P13/Q2/e

2(e)	1 mark for each correctly completed term; The bit depth of a bitmap image is the number of bits that are used to store each pixel. Metadata about the image is stored in the header of the file.	2
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49. O/N 22/P13/Q9/a,b

9(a)(i)	93	1
9(a)(ii)	147	1
9(b)	1 mark for each correct benefit (max 2) Examples: • straightforward to convert to / from BCD and denary • ... so it is less complex to encode and decode for programmers • easier for digital equipment use BCD to display output information • can represent monetary values exactly	2

50. O/N 21/P11/Q1/a,b,c

1(a)	1 mark for each correct line	5
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1(b)(i)	1 mark for answer 1 mark for working e.g. <pre> 1010 1010 0011 0111 <u>1110 0001</u> 1 1 1 1 1 1 </pre>	2
1(b)(ii)	The result is a larger number than can be stored in the given number of bits. // The result is greater than <u>255</u>	1
1(c)	240	1

51. O/N 21/P11/Q7/a,b

7(a)(i)	1 mark per bullet point • Smaller time gaps between the samples • Makes the digital sound wave more accurate • Smaller quantisation errors	2
7(a)(ii)	1 mark per bullet point • More samples/data are taken/recorded • ... so more bits are stored altogether	2
7(b)(i)	1 mark per bullet point • Reduces the file size • Faster to transmit/download • Original file is too large for email storage/attachment	2
7(b)(ii)	1 mark per bullet point to max 2 e.g. • Reduce amplitude to only the range used • ... limited amplitudes mean fewer bits per sample • Run-length-encoding • ... Where consecutive sounds are the same record the binary value of the sound and number of times it repeats • Record the changes instead of the actual sounds	2

52. O/N 21/P12/Q4/a,b,c,d,e

4(a)	205	1
4(b)	-51	1
4(c)	CD	1

4(d)	1 mark for: The denary value in each group of 4 bits is greater than 9 // the denary value in each nibble is greater than 9	1
4(e)(i)	1 mark for working, 1 mark for answer <pre> 0011 1101 +0010 1101 ----- 0110 1010 111 1 1 </pre>	2
4(e)(ii)	1 mark for working, 1 mark for answer <pre> 0011 1101 +1101 0011 (two's complement) ----- 0001 0000 1111 111 </pre>	2

53. O/N 21/P12/Q5/a,b

5(a)	1 mark for each description, 1 mark for each valid example <table border="1"> <thead> <tr> <th>Term</th><th>Description</th><th>Example from logo</th></tr> </thead> <tbody> <tr> <td>Property</td><td>data about the shapes // defines one aspect of the appearance of the drawing object</td><td>e.g. black line // white fill // black fill //solid (line) // font of letter // colour of triangle</td></tr> <tr> <td>Drawing list</td><td>the list of shapes involved in an image // a list that stores the command/description required to draw each object</td><td>e.g. triangle // capital letter R // rectangle // line</td></tr> </tbody> </table>	Term	Description	Example from logo	Property	data about the shapes // defines one aspect of the appearance of the drawing object	e.g. black line // white fill // black fill //solid (line) // font of letter // colour of triangle	Drawing list	the list of shapes involved in an image // a list that stores the command/description required to draw each object	e.g. triangle // capital letter R // rectangle // line	4
Term	Description	Example from logo									
Property	data about the shapes // defines one aspect of the appearance of the drawing object	e.g. black line // white fill // black fill //solid (line) // font of letter // colour of triangle									
Drawing list	the list of shapes involved in an image // a list that stores the command/description required to draw each object	e.g. triangle // capital letter R // rectangle // line									
5(b)(i)	1 mark for each bullet point to max 2 for each difference • Bitmap made up of pixels // bitmap is made of colours stored for individual pixels • Vector graphic store a set of instructions about how to draw the shape • When bitmap is enlarged the pixels get bigger and it pixelates • When vector is enlarged it is recalculated and does not pixelate • Bitmap files are usually bigger than vector graphics files because of the need to store data about each pixel • Vector graphics have smaller file size because they contain just the instructions to create the shapes • Bitmap images can be compressed with significant reduction in file size • Vector graphic images do not compress well because of little redundant data	4									

5(b)(ii)	1 mark for each bullet point to max 2 for each method • Reduce bit depth • ... reduces the number of bits per colour / pixel which means each pixel has fewer bits • Reduce colour palette // reduce number of colours • ... fewer colours mean fewer bits needed to store each colour • Reduce image resolution • ... fewer pixels per unit measurement means less binary to store	4
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54. O/N 21/P13/Q1/a,b,c

1(a)	1 mark for each correct line	5
1(b)(i)	1 mark for answer 1 mark for working e.g. <pre> 1010 1010 0011 0111 <u>1110 0001</u> 1 1 1 1 1 1 </pre>	2

1(b)(ii)	The result is a larger number than can be stored in the given number of bits. // The result is greater than <u>255</u>	1
1(c)	240	1

55. O/N 21/P13/Q7/a,b

7(a)(i)	1 mark per bullet point • Smaller time gaps between the samples • Makes the digital sound wave more accurate • Smaller quantisation errors	2
7(a)(ii)	1 mark per bullet point • More samples/data are taken/recorded • ... so more bits are stored altogether	2
7(b)(i)	1 mark per bullet point • Reduces the file size • Faster to transmit/download • Original file is too large for email storage/attachment	2
7(b)(ii)	1 mark per bullet point to max 2 e.g. • Reduce amplitude to only the range used • ... limited amplitudes mean fewer bits per sample • Run-length-encoding • ... Where consecutive sounds are the same record the binary value of the sound and number of times it repeats • Record the changes instead of the actual sounds	2

56. 9608 O/N 21/P11/Q1/a,b,c

1(a)(i)	An individual shape (in a vector graphic) // a shape defined using mathematics/geometry // a distinct element within an image // possible to alter without affecting the other objects within the image	1
1(a)(ii)	One mark per bullet point to max 4 E.g. • Line colour • Line width • Fill colour • Shape • Outline style • Dimensions / size • Position	4

1(b)	One mark per bullet point to max 3 E.g. • Dimensions e.g. 100 × 100 pixels // image size • File size • Colour_depth // bit depth • Location/offset of data within the file • Compression type • Confirmation that it is a bitmap // filetype • Image resolution • Colour palette	3
1(c)	One mark per bullet point to max 2 for each method. Lossy • Reduce the resolution • ...fewer pixels per unit measurement • ...fewer pixels / binary numbers are stored • Reduce the colour depth • ...reduce the number of bits per colour • ...each pixel has fewer bits Lossless • RLE (Run Length Encoding) • ...looks for runs of consecutive pixels of the same colour • ...stores the colour value once and the number of times it occurs	4

57. 9608 O/N 21/P11/Q2/a

2(a)	One mark per correct definition <table><tr><th>Term</th><th>Definition</th></tr><tr><td>Sampling</td><td>Measuring the amplitude of the wave at regular/set time intervals</td></tr><tr><td>Sampling resolution</td><td>The number of bits used to represent each sample</td></tr><tr><td>Sampling rate</td><td>The number of samples taken per unit of time</td></tr></table>	Term	Definition	Sampling	Measuring the amplitude of the wave at regular/set time intervals	Sampling resolution	The number of bits used to represent each sample	Sampling rate	The number of samples taken per unit of time	3
Term	Definition									
Sampling	Measuring the amplitude of the wave at regular/set time intervals									
Sampling resolution	The number of bits used to represent each sample									
Sampling rate	The number of samples taken per unit of time									

58. M/J 21/P11/Q1/a,b,c

1(a)(i)	1 mark for each description Pixel: • A single square of one colour • The smallest addressable element in an image File header: • Data about the bitmap image (e.g. number of colours)	2
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1(a)(ii)	1 mark per bullet point for working, 1 mark for answer Working: $1024 \times 512 = 524\,288$ pixels/bytes $524288 / 1024 / 1024$ Answer: 0.50 mebibytes	3
1(b)	1 mark for naming method, 1 mark per description to max 2 - Run-length encoding - Replace sequences of the same colour pixel ... with colour code and number of identical pixels	3
1(c)(i)	252	1
1(c)(ii)	1 mark per bullet point - Converting 15 to binary 0000 1111 - Method for addition - Final answer $\begin{array}{r} 0010\ 0011 \\ +\ 0000\ 1111 \\ \hline 0011\ 0010 \\ 1\ 111 \end{array}$	3
1(c)(iii)	1 mark per bullet point - Converting -10 to two's complement binary 1111 0110 - Adding values - Final answer 0001 1001 $\begin{array}{r} 10 = 0000\ 1010 \\ -10 = 1111\ 0110 \\ \hline 0010\ 0011 \\ +\ 1111\ 0110 \\ \hline 0001\ 1001 \\ 11\ 11 \end{array}$	3

59. M/J 21/P12/Q4/b(i)

4(b)(i)	102	1
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60. M/J 21/P12/Q6/a,b,c

6(a)	1 mark for each correct answer ASCII = $128 // 2^7$ Extended ASCII = $256 // 2^8$	2
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6(b)	1 mark per bullet point to max 2 - Each character has its own unique code - Each character in the word is replaced by its code - The codes are stored in the order in the word	2
6(c)(i)	31	1
6(c)(ii)	53	1

61. M/J 21/P13/Q1/a,b,c

1(a)(i)	1 mark for each description Pixel: - A single square of one colour - The smallest addressable element in an image File header: - Data about the bitmap image (e.g. number of colours)	2
1(a)(ii)	1 mark per bullet point for working, 1 mark for answer Working: $1024 \times 512 = 524\,288$ pixels/bytes $524288 / 1024 / 1024$ Answer: 0.50 mebibytes	3
1(b)	1 mark for naming method, 1 mark per description to max 2 - Run-length encoding - Replace sequences of the same colour pixel ... with colour code and number of identical pixels	3
1(c)(i)	252	1
1(c)(ii)	1 mark per bullet point - Converting 15 to binary 0000 1111 - Method for addition - Final answer 0010 0011 + 0000 1111 0011 0010 1 111	3

1(c)(iii)	1 mark per bullet point • Converting -10 to two's complement binary 1111 0110 • Adding values • Final answer 0001 1001 $ \begin{array}{r} 10 = 0000 \ 1010 \\ -10 = 1111 \ 0110 \\ \\ \begin{array}{r} 0010 \ 0011 \\ + \ 1111 \ 0110 \\ \hline 0001 \ 1001 \\ 11 \quad 11 \end{array} \end{array} $	3
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62. 9608 M/J 21/P11/Q2/a,b,c

2(a)	1 mark per bullet point to max 4 • The position/coordinate of the centre of the circle • Radius • Line width • Line colour • Line style • Fill colour	4
2(b)	1 mark per bullet point • A list that stores each separate object in the logo // a list that stores the command/description required to draw each object • Each shape has its own drawing list • Example related to logo, e.g. Three triangles, one square and one circle	2
2(c)	1 mark per drawback, 1 mark for expansion (max 2 each) • A bitmap file is likely to take up more storage space • ...because the colour of each pixel needs to be stored • A bitmap image cannot be enlarged // difficult to use in different types of document • ...without the image pixelating • A bitmap would be more difficult to edit • ...because each pixel would need to be edited separately	4

63. 9608 M/J 21/P11/Q3/c

3(c)(i)	1100 1001	1
3(c)(ii)	256	1

64. 9608 M/J 21/P11/Q8/b

8(b)(i)	1 mark per bullet point to max 2 - To reduce the storage space required to store the file - To reduce the time it would take to transmit the file - Because they have to email it and have limited file sizes for sending/receiving	2
8(b)(ii)	1 mark per bullet point - All of the original data needs to be retained/recoverable ... because otherwise the file contents would not make sense	2

65. 9608 M/J 21/P12/Q3/c

3(c)(i)	0011 1011	1
3(c)(ii)	3B	1

66. 9608 M/J 21/P12/Q7/b,c

7(b)	1 mark per bullet point to max 4 e.g. - Vertical height in pixels - Horizontal width in pixels - Bit depth/number of bits per pixel - Total file size in bytes - Offset in bytes where the image data is located - Type of compression used	4
7(c)	Justification 1 mark per bullet point to max 3 Lossy: - Lossy compression will achieve a much higher compression ratio (than lossless) for photographic images... ...so the images will load faster - There will be little identifiable change to the viewed images - It may not be important that the originals are not recoverable Lossless: - Images may be required in their original form // may need to be recovered completely - Images may already be a small file size so major reduction in file size is not needed... ...as the files will load quickly	3

67. 9608 M/J 21/P13/Q3/c

3(c)(i)	1110 1010	1
3(c)(ii)	EA	1

3(c)(iii)	–39	1
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68. 9608 M/J 21/P13/Q9/a

9(a)(i)	Sampling rate	1
9(a)(ii)	Sampling resolution // bit depth	1

69. 9608 M/J 21/P21/Q1/a

1(a)(i)	One mark for each part statement: • Each character is assigned... • ...a unique value • ...using 7 bits Max 2 marks	2												
1(a)(ii)	One mark for two correct, 2 marks for all correct <table><tr><th>Memory location</th><th>ASCII character value</th></tr><tr><td>100</td><td>70</td></tr><tr><td>101</td><td>65</td></tr><tr><td>102</td><td>68</td></tr><tr><td>103</td><td>69</td></tr><tr><td>104</td><td>68</td></tr></table>	Memory location	ASCII character value	100	70	101	65	102	68	103	69	104	68	2
Memory location	ASCII character value													
100	70													
101	65													
102	68													
103	69													
104	68													

70. 9608 M/J 21/P22/Q1/b

1(b)	One mark for: - Program / computer / CPU can only process / store <u>binary</u> values. - Standard encoding recognised by all programs / used to exchange information Note: Max 1 mark	1
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71. M/J 20/P11/Q4/a,b,c

4(a)	1 mark per bullet point $2000 * 1000 * 24 = 48\,000\,000$ bits $48\,000\,000 / 8 / 1024 / 1024$ $= 6$ MB or 5.7 MB	3
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4(b)	1 mark per bullet point to max 2 • Only 1 bit needed to store the colour of each pixel ... • ... so number of pixels * bit depth is 2000 * 1000 * 1 (rather than 2000 * 1000 * 24) • ... so the calculation (in part 4(a)) results in smaller figure for file size	2						
4(c)(i)	0110 0010	1						
4(c)(ii)	1 mark for each correct line <table><tr><td>Character</td><td>t</td></tr><tr><td>ASCII denary value</td><td>116</td></tr><tr><td>Hexadecimal value</td><td>74</td></tr></table>	Character	t	ASCII denary value	116	Hexadecimal value	74	2
Character	t							
ASCII denary value	116							
Hexadecimal value	74							

72. M/J 20/P11/Q9/b

9(b)	1 mark for 3 correct answers <table border="1"> <tr> <th>Action</th><th>Lossy</th><th>Lossless</th></tr> <tr> <td>Reducing the resolution of an image</td><td>✓</td><td></td></tr> <tr> <td>Using run-length encoding on a text file</td><td></td><td>✓</td></tr> <tr> <td>Reducing the sampling rate of a sound file</td><td>✓</td><td></td></tr> </table>	Action	Lossy	Lossless	Reducing the resolution of an image	✓		Using run-length encoding on a text file		✓	Reducing the sampling rate of a sound file	✓		1
Action	Lossy	Lossless												
Reducing the resolution of an image	✓													
Using run-length encoding on a text file		✓												
Reducing the sampling rate of a sound file	✓													

73. M/J 20/P12/Q1/b

1(b)(i)	1 mark per bullet point to max 3 - It is a series of (geometric) shapes / lines / objects - Stored coordinates - Contains a drawing list - Commands / formulae for creating each individual object - and the attributes/property for that object - E.g. the colour, thickness of line etc.	3
1(b)(ii)	1 mark per bullet point to max 2 - Needs to be large for the signs without becoming pixelated. - Smaller file size means faster transfer rates are possible. - Smaller file size reduces storage requirements when stored many times (on multiple documents).	2

74. M/J 20/P12/Q2/a,b

2(a)	1 mark for each correct term/definition		3
	Term	Definition	
	Sampling	The amplitude of the analogue sound wave is recorded at regular/set intervals	
	Sampling rate	The number of samples per unit time	
	Sampling resolution	The number of bits used to store each sample	
2(b)(i)	1 mark only from: - Run-length encoding - Huffman Coding - Any other valid encoding method		1
2(b)(ii)	1 mark per bullet point to max 2 - Reduce sampling rate ... fewer samples taken per second means less data is being stored Or - Reduce sample resolution ... fewer bits are used to represent each sample, so less data is stored		2

75. M/J 20/P12/Q5/b

5(b)(i)	11000010	1
5(b)(ii)	CD	1
5(b)(iii)	1 mark per bullet point to max 2 - The maximum range for an 8-bit two's complement binary number is –128 to +127 ... 200 is outside of the maximum range	2

76. M/J 20/P13/Q5/b

5(b)	1 mark for a lossless technique 1 mark per bullet point for justification to max 2 - Lossless does not lose any data - Any lost data will mean the program will not work	3
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77. M/J 20/P13/Q7/b,c,d

7(b)	1 mark per bullet point • 160 * 160 (= 25600 pixels) • 25600 * 3 (= 76800 bytes) • 75 KB (divide by 1024) or 76.8 KB (divide by 1000)	3												
7(c)	1 mark per bullet point to max 2 • The logo can be enlarged without becoming pixelated • because the instructions to create the logo are stored or • (Usually) smaller file size • only coordinates and calculations stored instead of individual pixels	2												
7(d)	1 mark for 1 or 2 correct answers 2 marks for 3 correct answers <table border="1"><thead><tr><th></th><th>Red</th><th>Green</th><th>Blue</th></tr></thead><tbody><tr><td>Hexadecimal value</td><td>91</td><td>3C</td><td>8E</td></tr><tr><td>Denary value</td><td>145</td><td>60</td><td>142</td></tr></tbody></table>		Red	Green	Blue	Hexadecimal value	91	3C	8E	Denary value	145	60	142	2
	Red	Green	Blue											
Hexadecimal value	91	3C	8E											
Denary value	145	60	142											

78. O/N 19/P11/Q5/a,b,c,d,e

5(a)(i)	256	1
5(a)(ii)	78	1
5(b)	1111 0011 1110	1
5(c)(i)	69	1
5(c)(ii)	Any scenario where a single digit needs to be transmitted / displayed e.g. calculator / digital clock	1
5(d)	1 mark per bullet to max 2 • Each character is represented by a unique • ... denary / hexadecimal / binary number	2
5(e)(i)	1 mark per bullet to max 2 Lossless: • All the data is needed // the original file is fully restored • If any data is lost, the program will not run • Probably does not require significant reduction in file size // a program written in a high-level language is just text, so does not need much reduction in size	2

5(e)(ii)	1 mark per bullet to max 2 Lossy: • All the data is not required • The number of colours / resolution can be reduced without the user noticing • Email requires a significantly smaller file size // takes less time to transmit Lossless: • A high quality image may be needed • All of the data is needed // cannot afford to lose any data // the original file is fully restored	2
5(e)(iii)	1 mark per bullet to max 2 Lossy: • Some loss of quality will not be noticed // high quality video not needed on the website • A more significant reduction may be needed • Takes a shorter time to upload / download // requires less bandwidth Lossless: • A high quality video may be needed • Might only be a short video clip • All of the data is needed // cannot afford to lose any data // the original file is fully restored	2

79. O/N 19/P12/Q6/d(i)

6(d)(i)	1 mark per bullet point to max 4 Max 3 for image Max 3 for sound Images • The images are stored as bitmaps • Each image is made up of pixels • ... each pixel is of a single colour • Each colour has a unique binary number • Store the sequence of binary numbers for each image / frame // store the binary value of each pixel Sound • Measure the height/amplitude of the sound wave • A set number of times per second // at regular time intervals • Each amplitude has a unique binary number • Store the sequence of binary numbers for each sample	4
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80. O/N 19/P13/Q2/b(i,ii)

2(b)(i)	1 mark per bullet point to max 2 • The amplitude of the wave is measured • ... at set, regular time intervals • The value is stored as a binary number	2
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2(b)(ii)	1 mark per bullet point - Sampling resolution of 44100 Hz takes more samples per second, so the file size will be larger // Sampling resolution of 21000 Hz takes fewer samples per second, so the file size will be smaller - At a resolution of 44100 Hz, the sound recording is a closer / more accurate representation of Leonardo's voice // At a resolution of 21000 Hz, the sound recording is a less accurate representation of Leonardo's voice	2
----------	--	----------

81. O/N 19/P13/Q4/a,b,c,d,e

4(a)	1532	1
4(b)	1111 0001 0001	1
4(c)	101	1
4(d)	65	1
4(e)	DE	1

82. M/J 19/P11/Q1/d

1(d)(i)	1 mark per bullet point to max 1 - The data files are very large - It would take a long time to send the uncompressed file // Compressed files will download faster - A higher bandwidth would be needed to transmit the uncompressed file	1
1(d)(ii)	1 mark per bullet point to max 1 - Data is lost - The decompressed file is not the same as the original	1
1(d)(iii)	1 mark per bullet point to max 3 - Lossy creates a smaller file than lossless // lossy compresses further than lossless - The recording of the concert is a large file size and needs <u>significant</u> reduction in size - Lossy removes detail which can be lost without people noticing - By example e.g. reduction in sound quality <u>will not be noticed</u>	3

83. M/J 19/P11/Q6/d

6(d)(i)	1 mark per bullet point to max 3 - The amplitude of the wave is measured /sound wave is sampled - At <u>set/regular</u> time intervals - Each sample is stored as a binary number - Samples are stored in order in a file	3
6(d)(ii)	1 mark per bullet point to max 1 for each Sample rate: - Increasing the sample rate means more samples per second hence more bits per second <u>and</u> larger file size - Decreasing the sample rate means fewer samples per second hence fewer bits per second <u>and</u> smaller file size Sample resolution: - A higher sampling resolution means more bits per sample <u>and</u> a larger file size - A lower sampling resolution means fewer bits per sample. a smaller file size	2

84. M/J 19/P12/Q3/d

3(d)(i)	1 mark for correct answer 0100 0001	1
3(d)(ii)	1 mark for correct answer 41	1
3(d)(iii)	1 mark for correct answer 0044	1

85. M/J 19/P13/Q5/a,c

5(a)	1 mark for each correctly linked box on the right <table><thead><tr><th>Maximum number of colours</th><th>Minimum number of bits</th></tr></thead><tbody><tr><td>68</td><td>1</td></tr><tr><td>256</td><td>2</td></tr><tr><td>127</td><td>3</td></tr><tr><td>2</td><td>7</td></tr><tr><td>249</td><td>8</td></tr><tr><td></td><td>9</td></tr></tbody></table>	Maximum number of colours	Minimum number of bits	68	1	256	2	127	3	2	7	249	8		9	3
Maximum number of colours	Minimum number of bits															
68	1															
256	2															
127	3															
2	7															
249	8															
	9															
5(c)	1 mark per bullet 88.2 kHz - The sound wave is sampled <u>88200</u> times per second 32 bits - Each sample is stored as a 32-bit binary number	2														

86. O/N 18/P11/Q1/b

1(b)	1 mark per bullet point - The sampling resolution number of bits used to store each <u>sample</u> - Increasing the (sampling) resolution means a larger file size // Decreasing the (sampling) resolution means a smaller file size - Increasing the (sampling) resolution gives a more accurate representation of the analogue sound // Decreasing the (sampling) resolution gives a less accurate representation of the analogue sound - Increasing the (sampling) resolution means a greater range of values can be stored // Decreasing the (sampling) resolution gives a smaller range of values that can be stored - Increasing the (sampling) resolution reduces the quantization errors // Decreasing the (sampling) resolution causes greater quantization errors	3
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87. O/N 18/P11/Q4/c

4(c)(i)	186	1
4(c)(ii)	BA	1
4(c)(iii)	-70	1

88. O/N 18/P12/Q1/a,b,c

1(a)(i)	2	1
1(a)(ii)	1 mark per bullet point - Number of pixels: $6 \times 6 // 36$ - Number of bits: Number of pixels (36) $\times 2 \dots$ $\dots = 72$ <u>bits</u> // 9 <u>bytes</u>	3
1(b)(i)	1 mark per bullet point - Number of pixels: $1000 \times 1000 // 1\,000\,000$ - Number of bytes: Number of pixels (1 000 000) $\times 2 // 2\,000\,000 //$ Number of bits: Number of pixels (1 000 000) $\times 16 // 16\,000\,000$ - Conversion to <u>megabytes</u> 2 (MB) // 1.91 (MB)	4
1(b)(ii)	1 mark per method correctly linked to its description max 3 1 mark for each compression type correctly linked to its method(s). max 2 Description Removes pixels Reduces number of pixels per inch Uses fewer bits per pixel Stores colour code and count of repetitions Method Crop the photograph Use run-length encoding Use fewer colours Compression type <u>Lossy</u> Lossless	5
1(c)	1 mark per bullet point. Max 2 marks for each reason. - Smaller file size - Can be transferred quicker/downloaded quicker - Enlarges without pixilation - Needs to be used on different screens / devices / resolutions	4

89. O/N 18/P12/Q3/c

3(c)(i)	193	1
3(c)(ii)	C1	1
3(c)(iii)	-63	1

90. O/N 18/P13/Q1/a,b,c

1(a)(i)	3	1
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1(a)(ii)	1 mark per bullet point - Number of pixels: $6 \times 6 = 36$ - Number of bits: Number of pixels $(36) \times 3$ 108 bits / 13.5 bytes	3															
1(b)(i)	1 mark per bullet point - Number of pixels: $50\,000 \times 50\,000 = 2\,500\,000\,000$ - Number of bytes: Number of pixels $(2\,500\,000\,000) \times \underline{4} //$ 10 000 000 000 // - Number of bits: Number of pixels $(2\,500\,000\,000) \times \underline{32} //$ 80 000 000 000 - Conversion to gigabytes 10 GB / 9.3 GB	4															
1(b)(ii)	1 mark for correct tick in each row. <table border="1" data-bbox="375 817 1232 1142"> <thead> <tr> <th>Compression method</th><th>Lossy</th><th>Lossless</th></tr> </thead> <tbody> <tr> <td>Cropping the image</td><td>✓</td><td></td></tr> <tr> <td>Reducing the resolution of the image</td><td>✓</td><td></td></tr> <tr> <td>Using run-length encoding (RLE)</td><td></td><td>✓</td></tr> <tr> <td>Reducing the colour depth of the image</td><td>✓</td><td></td></tr> </tbody> </table>	Compression method	Lossy	Lossless	Cropping the image	✓		Reducing the resolution of the image	✓		Using run-length encoding (RLE)		✓	Reducing the colour depth of the image	✓		4
Compression method	Lossy	Lossless															
Cropping the image	✓																
Reducing the resolution of the image	✓																
Using run-length encoding (RLE)		✓															
Reducing the colour depth of the image	✓																
1(c)	1 mark per bullet point to max 3 - Looks for runs of consecutive pixel of the same colour - Stores the colour value once and the number of times it occurs - Lossless method of compression - Reference to the given image in context	3															

91. O/N 18/P13/Q2/b(i,ii,iii)

2(b)(i)	242	1
2(b)(ii)	F2	1
2(b)(iii)	-14	1

92. M/J 18/P11/Q2/a,b,c,d

2(a)	1 mark per bullet, max 2 - Made up of pixels - Each pixel has one colour - Colour of each pixel stored as a binary number	2
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2(b)(i)	1 mark per bullet, max 2 - Each pixel requires only one bit (as there are only two colours) - Black represented by 1 and white by 0 (or vice versa) - Bits are stored for each pixel in sequence 11111 01010 01010 01010 01010	2
2(b)(ii)	1 mark for the explanation - Stores the colour and the number of times it occurs 1 mark for example from - An example from the bitmap given e.g. B5, W1, B1 and so on	2
2(c)	1 mark per bullet - Number of pixels $500 \times 1000 (= 500\,000)$ 35 colours require 6 bits per pixel - Number of bytes $(500\,000 \times 6) / 8 = 3\,000\,000 / 8 (= 375\,000)$ = 375 Kb	4
2(d)	1 mark per bullet to max 2 marks per benefit - Can resize it without pixilation - Image is redrawn/recalculated with each adjustment - Smaller file size - Storing points/equations/commands etc., not individual pixels	4

93. M/J 18/P11/Q8/c

8(c)(i)	135	1
8(c)(ii)	87	1
8(c)(iii)	-121	1

94. M/J 18/P12/Q4/a,b

4(a)(i)	219	1
4(a)(ii)	DB	1
4(a)(iii)	-37	1
4(b)(i)	1 mark from: - The <u>symbols</u> that the <u>computer</u> recognises/uses - A list of <u>characters</u> recognised by the <u>computer</u> hardware and software	1

4(b)(ii)	1 mark per bullet to max 2 • UNICODE has greater range of characters than ASCII • UNICODE represents most written languages in the world while ASCII does not ASCII used for English only • ASCII uses 7 or 8 bits or one byte whereas UNICODE uses up to 4 bytes per character • UNICODE is standardised while ASCII is not	2
4(b)(iii)	1 mark for correct working, 1 mark for correct answer Working: $\text{Code for Z} = \text{Code for A} + 25_{10}$ $\text{Code for Z} = 41_{16} + 25_{10}$ $\text{Code for Z} = 41_{16} + 19_{16}$ $\text{Code for Z} = 5A_{16}$ Answer: $5A_{16}$	2

95. M/J 18/P12/Q5/a,b,c

5(a)	1 mark per bullet to max 3 • Amplitude (of the sound wave) measured • At <u>set / regular</u> time intervals / per time unit / time period • Value of the sample is recorded as a binary number	3
5(b)	1 mark per bullet to max 2 • (Increasing the sampling resolution means) more bits per sample // larger range of values • Larger file size • More accurate representation of sound	2
5(c)	1 mark per bullet to max 3 • Fewer samples (per unit time) • File size will decrease • Larger gaps / spaces between samples // Greater quantization errors • Sound accuracy will reduce // not as close to original sound	3

96. M/J 18/P13/Q3/c

3(c)(i)	193	1
3(c)(ii)	C1	1
3(c)(iii)	- 63	1
3(c)(iv)	The <u>first 4 bits / first nibble</u> (would give 12 which) is <u>$\geq 9 / 2$ digits</u> (which is not valid for BCD)	1

97. M/J 18/P13/Q6/a,b,c,d

6(a)	1	1
6(b)	1 mark for correct method (colour code and number of pixels) 1 mark for first 7 groups correct 1 mark for remainder correct 3B9 1A3 3B3 1A2 3B1 1A2 3B2 1A1 3B3 1A1 3B2 1A2 3B1 1A2 3B3 1A3 3B9	3
6(c)	5	1
6(d)	1 mark for purpose - Stores data about the file contents/image/metadata Max 2 marks for examples of contents <u>Confirmation</u> that the file is a BMP // confirmation of file type - File size - Location / offset of image data within the file - Dimensions of the image (in pixels) // <u>image</u> resolution - Colour depth (bits per pixel, 1, 4, 8, 16, 24 or 32) - Type of compression used (if any)	3

98. O/N 17/P13/Q1/a,b

1(a)(i)	119	1								
1(a)(ii)	−120	1								
1(a)(iii)	<table><tr><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td></tr></table>	1	1	1	0	1	1	1	1	1
1	1	1	0	1	1	1	1			
1(a)(iv)	Lowest value: −128 Highest value: +127	1								
1(b)(i)	0110 0101 0011	1								
1(b)(ii)	The second block of four binary digits represents a digit larger than 9 // 14	1								
1(b)(iii)	A string of digits on any electronic device displaying numeric values	1								

99. O/N 17/P22/Q1/b(i)

1(b)(i)	<table><tr><th>Data</th></tr><tr><td></td></tr><tr><td>67 // 0100 0011 // 043h</td></tr><tr><td>65 // 0100 0001 // 041h</td></tr><tr><td>71 // 0100 0111 // 047h</td></tr><tr><td>69 // 0100 0101 // 045h</td></tr><tr><td></td></tr></table> One mark for 67 and 65 One mark for 71 and 69 Accept binary, denary or hex values (hex must be clearly indicated) Max one mark if blank cell anywhere in sequence Ignore any data values before or after the four characters	Data		67 // 0100 0011 // 043h	65 // 0100 0001 // 041h	71 // 0100 0111 // 047h	69 // 0100 0101 // 045h		2
Data									
67 // 0100 0011 // 043h									
65 // 0100 0001 // 041h									
71 // 0100 0111 // 047h									
69 // 0100 0101 // 045h									

100. M/J 17/P11/Q3/a,b

3(a)	<i>Sampling rate</i> The <u>number of samples</u> taken <u>per unit time</u> // the number of times the amplitude is measured <u>per unit time</u> 1 Increasing the sampling rate will increase the accuracy / precision of the digitised sound // Increasing the sampling rate will result in smaller quantisation errors. 1	2
3(b)(i)	<i>Pixel</i> Smallest picture element which can be drawn 1 <i>Screen resolution</i> The number of pixels which can be viewed horizontally and vertically on the screen // or by example - A typical screen resolution is 1680 pixels × 1080 pixels. 1	2
3(b)(ii)	8	1
3(b)(iii)	<i>Working: Max two from:</i> - Number of pixels is 2048×512 1 - One pixel will be stored as one byte 1 - Number of kilobytes = $(2048 \times 512) / 1024$ 1 <i>Answer: One mark:</i> Number of kilobytes = 1024 KB 1	3
3(b)(iv)	One from: - Confirmation that the file is a BMP 1 - File size 1 - Location/offset of image data within the file 1 - Dimensions of the image in pixels // image resolution 1 - Colour depth (bits per pixel) 1 - Type of compression used, if any 1	1

101. M/J 17/P11/Q4/d(i)

4(d)(i)	07 C2		2
	07	1	
	C2	1	

102. M/J 17/P12/Q3/a,b

3(a)	Definition: Max two from: - The number of distinct values available to encode/represent each sample 1 - Specified by the number of bits used to encode the data for one sample 1 - Sometimes referred to as bit depth 1 Explanation: Max two from: - A larger sampling resolution will mean there are more values available to store each sample 1 - A larger sampling resolution will improve the accuracy of the digitised sound // A larger sampling resolution will decrease the distortion of the sound 1 - Increased sampling resolution means a smaller quantization error 1		Max 3
3(b)(i)	One from: - The <u>number of pixels</u> per <u>unit measurement</u> 1 - The number of pixels in an image 1 - The number of pixels wide by the number of pixels high 1 - Number of pixels per row by the number of rows 1		1
3(b)(ii)	4		1
3(b)(iii)	Working: Max two from: - Number of pixels is 8192×256 1 - One pixel will be stored as one byte 1 - Number of kilobytes = $(8192 \times 256) / 1024$ 1 Answer: One mark: Number of kilobytes = 2048 KB 1		3
3(b)(iv)	Two from: - Confirmation that the file is a BMP 1 - File size 1 - Location/offset of image data within the file 1 - Dimensions of the image (in pixels) // image resolution 1 - Colour depth (bits per pixel, 1, 4, 8, 16, 24 or 32) 1 - Type of compression used, if any 1		Max 2

103. M/J 17/P12/Q5/c(i)

5(c)(i)	14 5E		2
	14	1	
	5E	1	

104. M/J 17/P13/Q3/a,b

3(a)	Sampling rate The <u>number of samples</u> taken <u>per unit time</u> // the number of times the amplitude is measured <u>per unit time</u> 1 Increasing the sampling rate will increase the accuracy / precision of the digitised sound // Increasing the sampling rate will result in smaller quantisation errors. 1	2
3(b)(i)	Pixel Smallest picture element which can be drawn 1 Screen resolution The number of pixels which can be viewed horizontally and vertically on the screen // or by example - A typical screen resolution is 1680 pixels × 1080 pixels. 1	2
3(b)(ii)	8	1
3(b)(iii)	Working: Max <u>two</u> from: - Number of pixels is 2048 × 512 1 - One pixel will be stored as one byte 1 - Number of kilobytes = (2048 × 512) / 1024 1 Answer: <u>One</u> mark: Number of kilobytes = 1024 KB 1	3
3(b)(iv)	<u>One</u> from: - Confirmation that the file is a BMP 1 - File size 1 - Location/offset of image data within the file 1 - Dimensions of the image in pixels // image resolution 1 - Colour depth (bits per pixel) 1 - Type of compression used, if any 1	1

105. M/J 17/P13/Q4/d(i)

4(d)(i)	07 C2	2
	07 1	
	C2 1	

106. O/N 16/P11/Q3/a,b

3 (a) (i) 0 0 1 0 1 1 1 0 [1]

(ii) 1 1 0 1 0 0 1 0 [1]

(iii) 2 E [1]

(b) (i) **One mark** for the explanation and **one mark** for the example

- Each denary digit is written as a 4-bit binary number
- Example: 46 = 0100 0110 [2]

(ii) **One mark** for the explanation and **one mark** for the example

- Binary number is split up into groups of 4 bits (starting from the right)
// Each group of 4 bits is converted to a denary digit
- Example: 0011 0111 = 37

[2]

107. O/N 16/P11/Q7/b

(b) **Four** from:

- Bitmap is made up of pixels
// Vector graphic store a set of instructions about how to draw the shape
- Bitmap files are usually bigger than vector graphics files // Take up more memory space
- Enlarging a bitmap can mean the image is pixelated
// vector graphic can be enlarged without the image becoming pixelated
- Bitmap images can be compressed (with significant reduction in file size)
// Vector graphic images do not compress well
- Bitmaps are suitable for photographs / scanned images
// Vector graphics are suitable for more geometric shapes
- Bitmap graphics use less processing power than vector graphics
- Individual elements of a bitmap cannot be grouped
// Individual elements of a vector graphic can be grouped
- Vector graphics need to be 'rasterised' in order to display or print

[4]

108. O/N 16/P11/Q8/c

(c) (i) Any **two** from:

- Only 128 / 256 characters can be represented
- Uses values 0 to 127 (or 255 if extended form) / one byte
- Many characters used in other languages cannot be represented
- In extended ASCII the characters from 128 to 255 may be coded differently in different systems

[2]

(ii) Any **two** from:

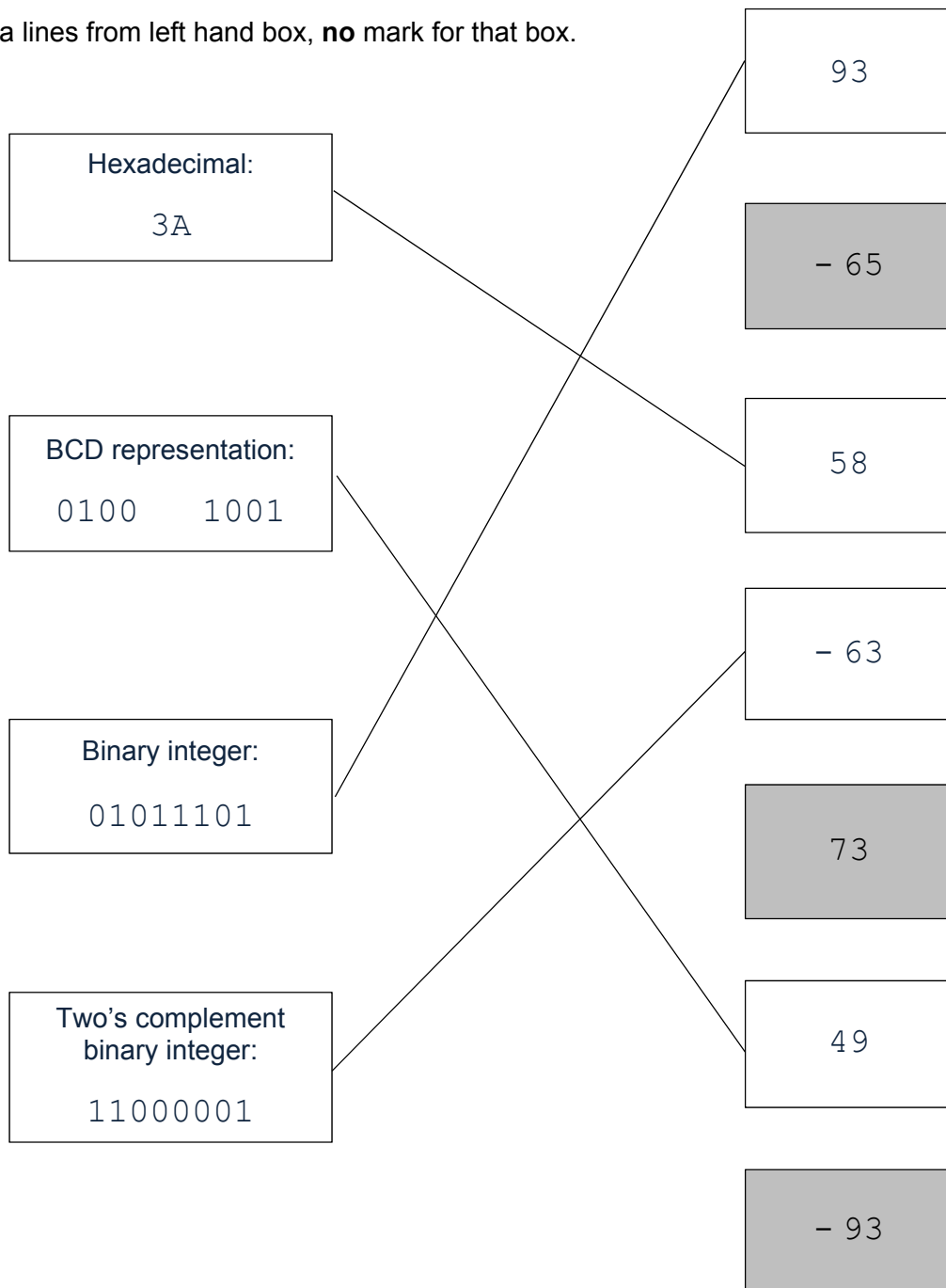
- Uses 16, 24 or 32 bits / two, three or four bytes
- Unicode is designed to be a superset of ASCII
- Designed so that most characters (in other languages) can be represented

[2]

109. O/N 16/P12/Q4

4 ONE mark for each correct line.

Extra lines from left hand box, **no** mark for that box.



110. O/N 16/P13/Q3/a,b

3 (a) (i) 0 0 1 0 1 1 1 0 [1]

(ii) 1 1 0 1 0 0 1 0 [1]

(iii) 2 E [1]

(b) (i) One mark for the explanation and **one mark** for the example

- Each denary digit is written as a 4-bit binary number
- Example: 46 = 0100 0110 [2]

(ii) One mark for the explanation and **one mark** for the example

- Binary number is split up into groups of 4 bits (starting from the right)
// Each group of 4 bits is converted to a denary digit
- Example: 0011 0111 = 37 [2]

111. O/N 16/P13/Q7/b

(b) Four from:

- Bitmap is made up of pixels
// Vector graphic store a set of instructions about how to draw the shape
- Bitmap files are usually bigger than vector graphics files // Take up more memory space
- Enlarging a bitmap can mean the image is pixelated
// vector graphic can be enlarged without the image becoming pixelated
- Bitmap images can be compressed (with significant reduction in file size)
// Vector graphic images do not compress well
- Bitmaps are suitable for photographs / scanned images
// Vector graphics are suitable for more geometric shapes
- Bitmap graphics use less processing power than vector graphics
- Individual elements of a bitmap cannot be grouped
// Individual elements of a vector graphic can be grouped
- Vector graphics need to be 'rasterised' in order to display or print [4]

112. O/N 16/P13/Q8/c

(c) (i) Any **two** from:

- Only 128 / 256 characters can be represented
- Uses values 0 to 127 (or 255 if extended form) / one byte
- Many characters used in other languages cannot be represented
- In extended ASCII the characters from 128 to 255 may be coded differently in different systems

[2]

(ii) Any **two** from:

- Uses 16, 24 or 32 bits / two, three or four bytes
- Unicode is designed to be a superset of ASCII
- Designed so that most characters (in other languages) can be represented

[2]

113. M/J 16/P11/Q2/a,b,c,d

2 (a) 00110111 [1]

(b) 83 [1]

(c) 10011010 [2]

Marks allocated as follows:

1 mark for the most significant bit

1 mark for the remaining 7 bits

(d) 78 [1]

114. M/J 16/P11/Q4/a,b

4 (a) Three from: [3]

- The height/amplitude of the (sound) wave is determined.
- At set (time) intervals // by example of sensible time period.
- To get an approximation of the sound wave
- And encoded as a sequence of binary numbers // and converted to a digital signal.
- Increasing the sampling rate will improve the accuracy of the recording.

(b) (i) No mark awarded for identifying method. **Three marks** for justification. [3]

Lossy – Three points from:

- The human ear will not notice that the decompressed stream will not be identical to the original (file) / that parts of the original data have been discarded / removed / deleted.
- File size reduction is greater than using lossless.
- Email has limits on file sizes (on attachments) / a smaller file will take less time to transmit.
- The file may not need to be of high precision / accuracy.
- The producer has requested an mp3 file.

Lossless – Three points from:

- The file needs to be high precision / accuracy.
- None of the original data is lost / the decompressed file will be identical to the original.
- The producer has requested a flac file.

(ii) **Three** points from:

[3]

- Lossless method of compression.
- Reduces (the physical size of) a string of adjacent, identical characters/pixels / bytes etc..
- The repeating string (a run) is encoded into two values.
- One value represents the number of (identical) characters in the run (the run count).
- The other value is the code of the character / colour code of pixel etc. in the run (the run value).
- The run value and run count combination may be preceded by a control character.
- Any valid example given.

(iii) **Two marks** for **three** correct rows, **one mark** for **two** correct rows.

[2]

Row 1: 153 10 255 3 153 3

Row 2: 153 9 255 6 153 1

Row 3: 153 7 255 9

Alternative correct answer:

Row 1: 153 9 255 2 153 2

Row 2: 153 8 255 5 153 0

Row 3: 153 6 255 8

115. M/J 16/P12/Q2/a,b,c,d

2 (a) 00110111

[1]

(b) 83

[1]

(c) 10011010

[2]

Marks allocated as follows:

1 mark for the most significant bit

1 mark for the remaining 7 bits

(d) 78

[1]

116. M/J 16/P12/Q4/a,b**4 (a) Three** from:**[3]**

- The height/amplitude of the (sound) wave is determined.
- At set (time) intervals // by example of sensible time period.
- To get an approximation of the sound wave
- And encoded as a sequence of binary numbers // and converted to a digital signal.
- Increasing the sampling rate will improve the accuracy of the recording.

(b) (i) No mark awarded for identifying method. **Three marks** for justification.**[3]****Lossy – Three** points from:

- The human ear will not notice that the decompressed stream will not be identical to the original (file) / that parts of the original data have been discarded / removed / deleted.
- File size reduction is greater than using lossless.
- Email has limits on file sizes (on attachments) / a smaller file will take less time to transmit.
- The file may not need to be of high precision / accuracy.
- The producer has requested an mp3 file.

Lossless – Three points from:

- The file needs to be high precision / accuracy.
- None of the original data is lost / the decompressed file will be identical to the original.
- The producer has requested a flac file.

(ii) Three points from:**[3]**

- Lossless method of compression.
- Reduces (the physical size of) a string of adjacent, identical characters/pixels / bytes etc..
- The repeating string (a run) is encoded into two values.
- One value represents the number of (identical) characters in the run (the run count).
- The other value is the code of the character / colour code of pixel etc. in the run (the run value).
- The run value and run count combination may be preceded by a control character.
- Any valid example given.

(iii) Two marks for **three** correct rows, **one mark** for **two** correct rows.**[2]**

Row 1: 153 10 255 3 153 3

Row 2: 153 9 255 6 153 1

Row 3: 153 7 255 9

Alternative correct answer:

Row 1: 153 9 255 2 153 2

Row 2: 153 8 255 5 153 0

Row 3: 153 6 255 8

117. M/J 16/P13/Q2/a,b,c,d

2 (a) 77 [1]

(b) 1000 0010 [1]

(c) -53 [2]

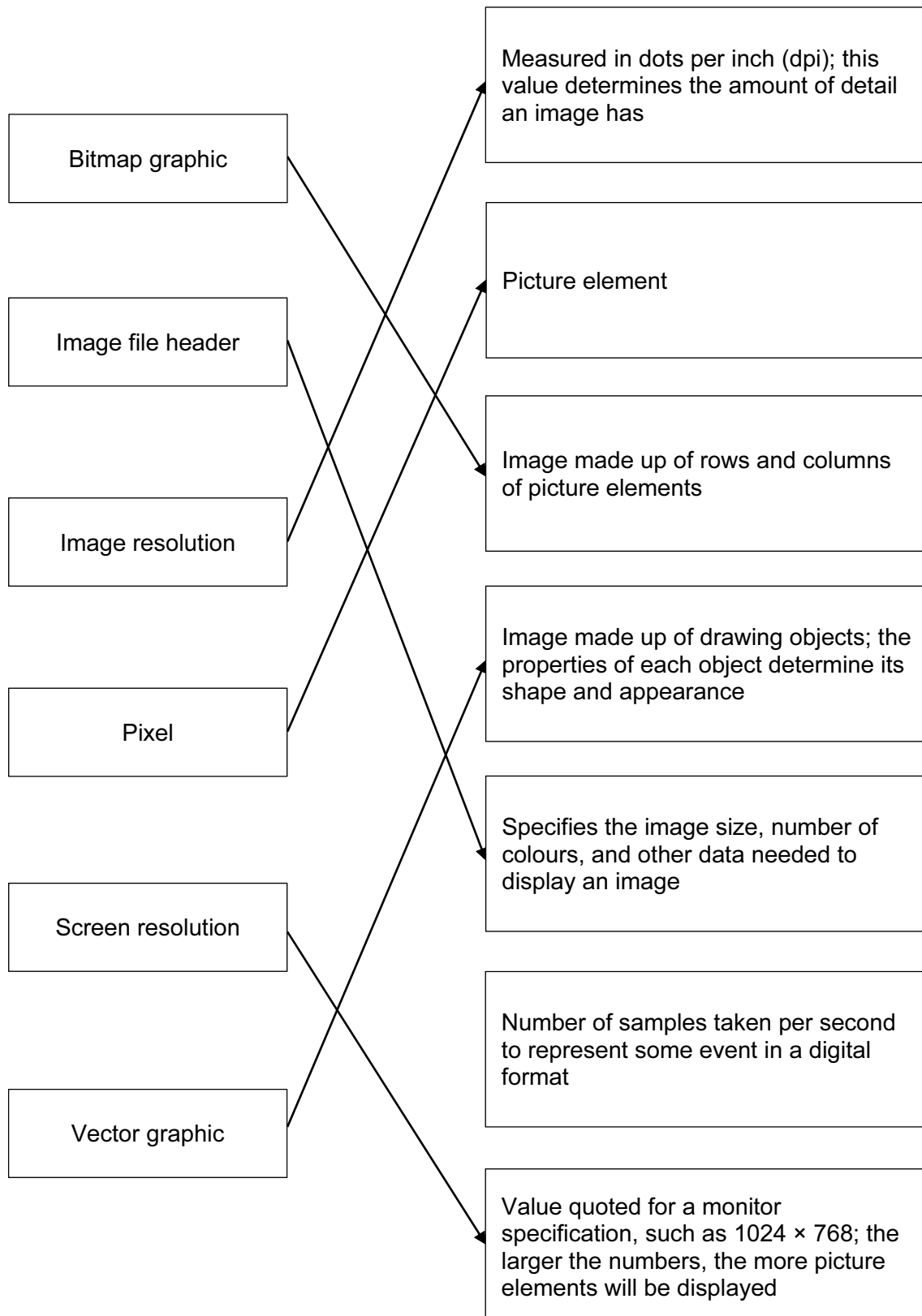
One mark for '53' and **one** mark for '-'

(d) C6 [2]

One mark for the answer, one mark for the method

- Working e.g. $198 / 16 = 12$, $198 - (12 \times 16) = 6$

118. O/N 15/P11/Q8/a,b

8 (a)

1 mark for each correct line, two lines from one box is incorrect

[6]

(b) (i) $\frac{512 \times 256}{8 \times 1024} = 16 \text{ KB}$

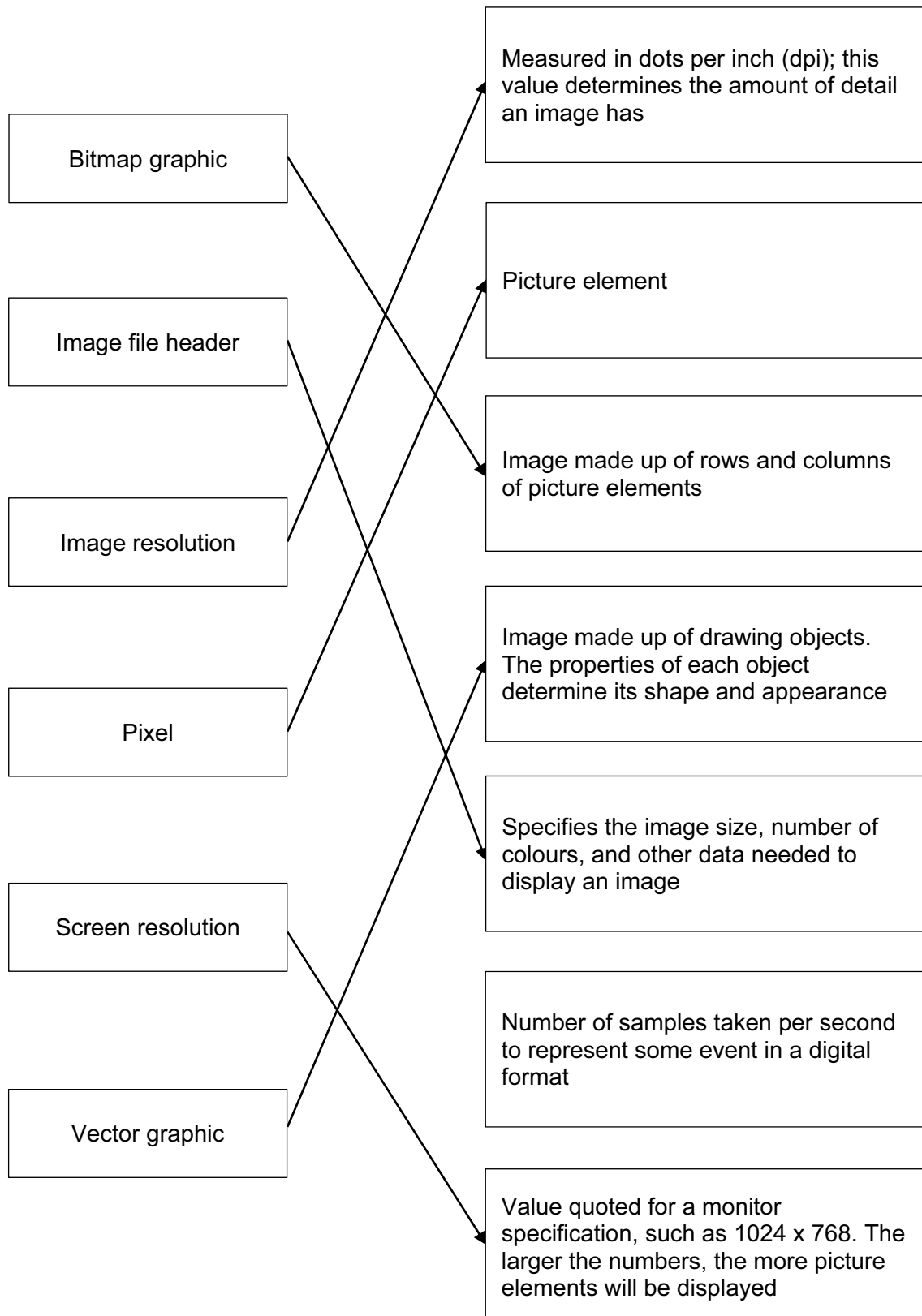
1 mark for numerator + 1 mark for denominator

[2]

- (ii) so it is possible to estimate how many images can be stored / to decide if it can be sent as an email attachment

[1]

119. O/N 15/P13/Q8/a,b

8 (a)

1 mark for each correct line, two lines from one box is incorrect

[6]

(b) (i) $\frac{512 \times 256}{8 \times 1024} = 16 \text{ KB}$

1 mark for numerator + 1 mark for denominator

[2]

- (ii) so it is possible to estimate how many images can be stored / to decide if it can be sent as an email attachment [1]

120. M/J 15/P11/Q1

(ii) 1 0 0 1 0 1 1 1

(iii)

114	0	1	1	1	0	0	1	0
– 93	1	0	1	0	0	0	1	1

[2]

1 (i) B 8

121. M/J 15/P11/Q2/a,c

2 (a) (i) Any **one** from:

- amplitude of sound wave taken at different points in time
- measurement of value of analogue signal at regular time intervals/a point in time [1]

(ii) Any **one** from:

- bit depth/sampling resolution sufficient for good quality sound
- higher bit depth/sampling resolution would mean bigger files
- ...hence less (music) content on each CD
- can represent dynamic range of about 90 dB
- 90 dB is basically the maximum dynamic range of human hearing
- compromise between quality and reasonable file size

[1]

(iii) Any **two** from:

- resolution is the number of distinct values available to encode/represent each sample
- specified by the number of bits used to store/record each sample
- sometimes referred to as bit depth
- the higher the sampling resolution, the smaller the quantization error
- a higher sampling resolution results in less distortion of the sound
- usually 8 bit, 16 bit, 24 bit or 32 bit

[2]

(iv) 1 mark for benefit and **1 mark** for drawback.

benefit

- allows for larger dynamic ranges
- ...as dynamic range is approximately six times the bit depth
- more accurate representation/crisper sound quality

drawback

- bigger files/occupies more memory/storage
- longer to transmit data/download music
- greater processing power needed

[2]



(c) Any **three** from:

For full marks both techniques must be mentioned.

- lossless designed to lose none of the original detail/lossless allows original file to be recreated exactly
- lossless technique based on some form of replacement
- mention of type of replacement, for example RLE, FLAC etc.
- by example: e.g. 000–1111–222222–333 = 3–0, 4–1, 6–2, 3–3 etc.
- maximum compression about 50%
- lossy may result in loss of detail compared to original file/lossy does not allow original file to be re-created exactly
- lossy techniques make decision about what parts of sound/sound file are important and discards other information
- only keeps sounds human ear can process/discards sounds most people cannot hear
- ... then applies lossless technique, for further reduction
- lossy compression can reduce to about 10%
- an example of jpeg, mp3 or other correct examples of compressed formats.

No double credit to opposite answers, e.g. lossless maintains detail, but lossy loses detail just one mark.

[3]

122. M/J 15/P12/Q1

(ii) 1 0 0 1 0 1 1 1

(iii)

114	0	1	1	1	0	0	1	0
– 93	1	0	1	0	0	0	1	1

[2]

1 (i) B 8



123. M/J 15/P12/Q2/a,c

2 (a) (i) Any **one** from:

- amplitude of sound wave taken at different points in time
- measurement of value of analogue signal at regular time intervals/a point in time [1]

(ii) Any **one** from:

- bit depth/sampling resolution sufficient for good quality sound
- higher bit depth/sampling resolution would mean bigger files
- ...hence less (music) content on each CD
- can represent dynamic range of about 90 dB
- 90 dB is basically the maximum dynamic range of human hearing
- compromise between quality and reasonable file size [1]

(iii) Any **two** from:

- resolution is the number of distinct values available to encode/represent each sample
- specified by the number of bits used to store/record each sample
- sometimes referred to as bit depth
- the higher the sampling resolution, the smaller the quantization error
- a higher sampling resolution results in less distortion of the sound
- usually 8 bit, 16 bit, 24 bit or 32 bit [2]

(iv) **1 mark** for benefit and **1 mark** for drawback.

benefit

- allows for larger dynamic ranges
- ...as dynamic range is approximately six times the bit depth
- more accurate representation/crisper sound quality

drawback

- bigger files/occupies more memory/storage
- longer to transmit data/download music
- greater processing power needed [2]



(c) Any **three** from:

For full marks both techniques must be mentioned.

- lossless designed to lose none of the original detail/lossless allows original file to be recreated exactly
- lossless technique based on some form of replacement
- mention of type of replacement, for example RLE, FLAC etc.
- by example: e.g. 000–1111–22222–333 = 3–0, 4–1, 6–2, 3–3 etc.
- maximum compression about 50%
- lossy may result in loss of detail compared to original file/lossy does not allow original file to be re-created exactly
- lossy techniques make decision about what parts of sound/sound file are important and discards other information
- only keeps sounds human ear can process/discards sounds most people cannot hear
- ... then applies lossless technique, for further reduction
- lossy compression can reduce to about 10%
- an example of jpeg, mp3 or other correct examples of compressed formats.

No double credit to opposite answers, e.g. lossless maintains detail, but lossy loses detail just one mark.

[3]

124. M/J 15/P13/Q1/a,b

1 (a) (i)

124	0	1	1	1	1	1	0	
–77	1	0	1	1	0	0	1	1

[2]

(ii) 124: 7 C

–77: B 3

[2]

(b) (i) 0 0 1 1 0 1 0 1 1 0 0 1 [1]

- (ii)**
- when denary numbers need to be electronically coded
 - e.g. to operate displays on a calculator where each digit is represented
 - decimal fractions can be accurately represented
- [2]